1. Project Overview

Overview

The Electronics Supply Chain Analytics project aims to analyze transactional and operational data to evaluate the overall performance of an electronics supply chain. The project applies data analysis techniques to transform raw operational data into meaningful insights that support better business decision-making across multiple functional areas.

The dataset contains information related to customer orders, products, inventory, suppliers, shipping operations, sales performance, and customer experience. By analyzing these interconnected components, the project provides a comprehensive understanding of supply chain operations and identifies key performance indicators that influence business efficiency and profitability.

The analytical process follows a structured workflow, beginning with data loading, exploration, cleaning, and validation, followed by exploratory data analysis (EDA). Business metrics are then calculated and organized into domain-specific analyses to evaluate commercial performance, inventory management, supplier efficiency, and customer satisfaction.

Project Objectives

The primary objectives of this project are to:

- Analyze the overall commercial performance of the business.
- Evaluate revenue, profit, and profitability across different business dimensions.
- Monitor inventory availability and identify potential stock-related risks.
- Assess supplier performance based on reliability, quality, and production metrics.
- Analyze customer purchasing behavior and satisfaction.
- Evaluate shipping efficiency and delivery performance.
- Identify operational patterns and business trends using exploratory data analysis.
- Generate data-driven insights that support operational improvements and strategic decision-making.

Project Workflow

The project follows the standard data analysis lifecycle:

1. **Data Collection**
2. **Data Understanding**

3. **Data Cleaning & Preprocessing**
4. **Data Validation**
5. **Exploratory Data Analysis (EDA)**
6. **Domain-Based Business Analysis**
    - Commercial Insights
    - Customer Experience Analysis
    - Supplier Performance Analysis
    - Inventory Analysis
    - Shipping & Delivery Analysis

Expected Outcomes

By the end of this project, the analysis provides:

- A cleaned and validated dataset suitable for analysis.
- Business performance indicators across multiple operational domains.
- Aggregated analytical results for each business area.
- Actionable insights that highlight strengths, weaknesses, and opportunities for improving supply chain performance.

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2. Dataset Description

2.1 Dataset Overview

The dataset used in this project represents the operational activities of an electronics supply chain. It contains transactional records that capture various aspects of customer orders, product information, inventory status, supplier performance, shipping operations, sales activities, and customer experience. Each record corresponds to a single customer order and provides detailed information that enables comprehensive business analysis across multiple operational domains.

The dataset consists of **32 features (columns) & 1500 row** describing different business entities and performance indicators. These features include order details, product specifications, pricing information, discounts, sales performance, supplier quality metrics, inventory management, logistics operations, customer demographics, and return information. Together, these attributes provide a comprehensive view of the end-to-end supply chain process and support data-driven decision-making.

To facilitate better understanding of the dataset, the following table provides a description of each feature, including its data type and business meaning.

To facilitate a better understanding of the dataset, The following table presents a detailed description of all dataset features, including their data types and business definitions.

Column Name	**Data Type**	**Description**
Order ID	Categorical	Unique identifier assigned to each customer order.
Order Date	Date	Date when the customer placed the order.
Ship Date	Date	Date when the order was shipped from the warehouse.
Delivery Date	Date	Date when the order was successfully delivered to the customer.
Product type	Categorical	Category of the product ordered (e.g., smartphones, laptops, tablets).
SKU	Categorical	Stock Keeping Unit (SKU) used to uniquely identify each product.
Category	Categorical	Product pricing segment or classification (e.g., budget, mid-range, premium).
Price	Numerical	Selling price of a single product unit before applying discounts.
Discount (%)	Numerical	Percentage discount applied to the product price.
Number of products sold	Numerical	Total quantity of products sold in the order.
Units Returned	Numerical	Number of units returned by the customer.
Return Reason	Categorical	Reason provided for returning the product.
Revenue generated	Numerical	Total revenue generated from the order after applying the discount.

Profit	Numerical	Net profit earned from the order.
Customer Satisfaction Score	Numerical	Customer satisfaction rating for the completed order.
Customer demographics	Categorical	Customer demographic group associated with the order.
Sales Channel	Categorical	Sales channel through which the order was placed (e.g., online, retail, wholesale).
Payment Method	Categorical	Payment method used to complete the purchase.
Region	Categorical	Geographic region where the order was placed or delivered.
Stock levels	Numerical	Available inventory quantity for the product at the time of the order.
Reorder Point	Numerical	Inventory threshold at which replenishment should be initiated.
Stockout Risk	Categorical	Indicates the likelihood of inventory shortage for the product.
Lead time	Numerical	Number of days required to replenish inventory from the supplier.
Shipping times	Numerical	Total number of days taken to ship the order.
Shipping costs	Numerical	Cost incurred for shipping the order.
On-Time Delivery	Categorical	Indicates whether the order was delivered on time (Yes/No).
Supplier name	Categorical	Name of the supplier responsible for providing the product.
Supplier Reliability Score	Numerical	Performance score representing the supplier's reliability.
Manufacturing costs	Numerical	Total manufacturing cost associated with the product.

Quality Score	Numerical	Quality evaluation score assigned to the product or supplier.
Defect rates	Numerical	Percentage of defective products associated with the supplier or product.
Production volumes	Numerical	Total number of units produced during the production cycle.

Summary

The dataset combines transactional, operational, and customer-related information into a single integrated data source. These features collectively support comprehensive analyses of sales performance, inventory management, supplier efficiency, logistics operations, and customer behavior. Understanding the purpose of each feature is essential before proceeding with data exploration, preprocessing, and analytical modeling.

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3. Data Analysis Workflow

Python Data Analysis Workflow

The data analysis process followed a structured workflow to ensure that the dataset was properly prepared before extracting business insights. The workflow began by loading the dataset into Python, followed by understanding its structure through exploratory inspection. Data preprocessing and cleaning techniques were then applied to improve data quality and remove inconsistencies. After validating the cleaned dataset, exploratory data analysis (EDA) was conducted to calculate key business metrics and generate analytical outputs for each business domain.

The workflow consisted of the following stages:

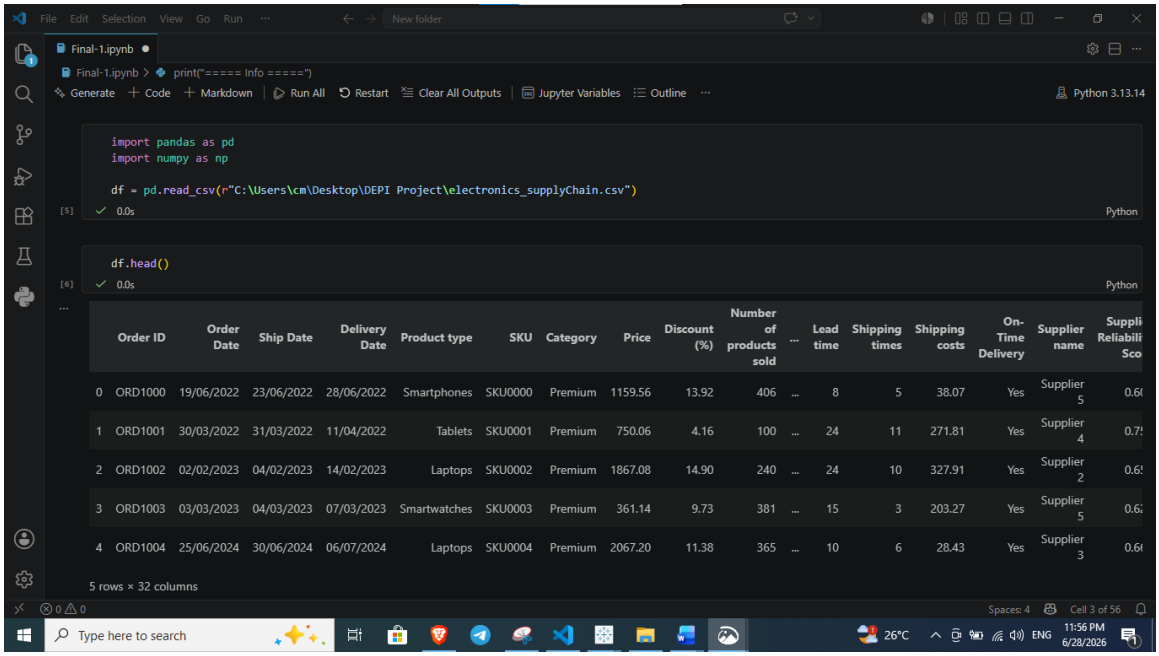
- Data Collection and Import
- Data Understanding
- Data Cleaning & Preprocessing
- Data Validation
- Exploratory Data Analysis (EDA)
- Domain-Based Business Analysis

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4. Data Collection and Import

The required Python libraries were imported, and the dataset was loaded into a pandas DataFrame using pd.read_csv(). An initial preview of the dataset was then displayed using df.head() to verify that the data had been loaded correctly and to inspect the first few records.

Output:



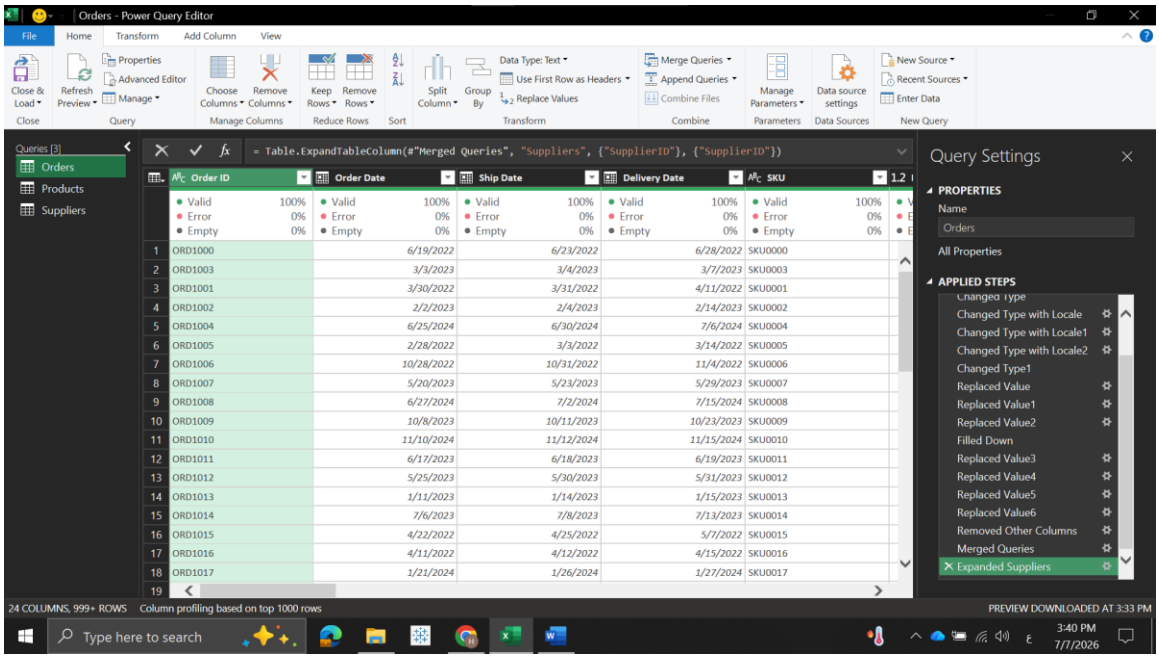
```
import pandas as pd
import numpy as np

df = pd.read_csv(r"C:\Users\cm\Desktop\DEPI Project\electronics_supplyChain.csv")
```

```
df.head()
```

	Order ID	Order Date	Ship Date	Delivery Date	Product type	SKU	Category	Price	Discount (%)	Number of products sold	...	Lead time	Shipping times	Shipping costs	On-Time Delivery	Supplier name	Suppli Reliabili Sco
0	ORD1000	19/06/2022	23/06/2022	28/06/2022	Smartphones	SKU0000	Premium	1159.56	13.92	406	...	8	5	38.07	Yes	Supplier 5	0.6(
1	ORD1001	30/03/2022	31/03/2022	11/04/2022	Tablets	SKU0001	Premium	750.06	4.16	100	...	24	11	271.81	Yes	Supplier 4	0.7
2	ORD1002	02/02/2023	04/02/2023	14/02/2023	Laptops	SKU0002	Premium	1867.08	14.90	240	...	24	10	327.91	Yes	Supplier 2	0.6
3	ORD1003	03/03/2023	04/03/2023	07/03/2023	Smartwatches	SKU0003	Premium	361.14	9.73	381	...	15	3	203.27	Yes	Supplier 5	0.6
4	ORD1004	25/06/2024	30/06/2024	06/07/2024	Laptops	SKU0004	Premium	2067.20	11.38	365	...	10	6	28.43	Yes	Supplier 3	0.6(

5 rows × 32 columns



Observation:

The dataset was successfully loaded and contains the expected columns representing order information, products, pricing, inventory, suppliers, shipping, and customer-related attributes.

- The dataset was successfully loaded into a pandas DataFrame and Power query.
- The dataset contains transactional records, where each record represents a customer order.
- The available attributes provide comprehensive information that supports subsequent data cleaning, validation, and exploratory data analysis.
- The initial preview confirmed that the dataset structure aligns with the project requirements and is ready for the next stage of analysis.

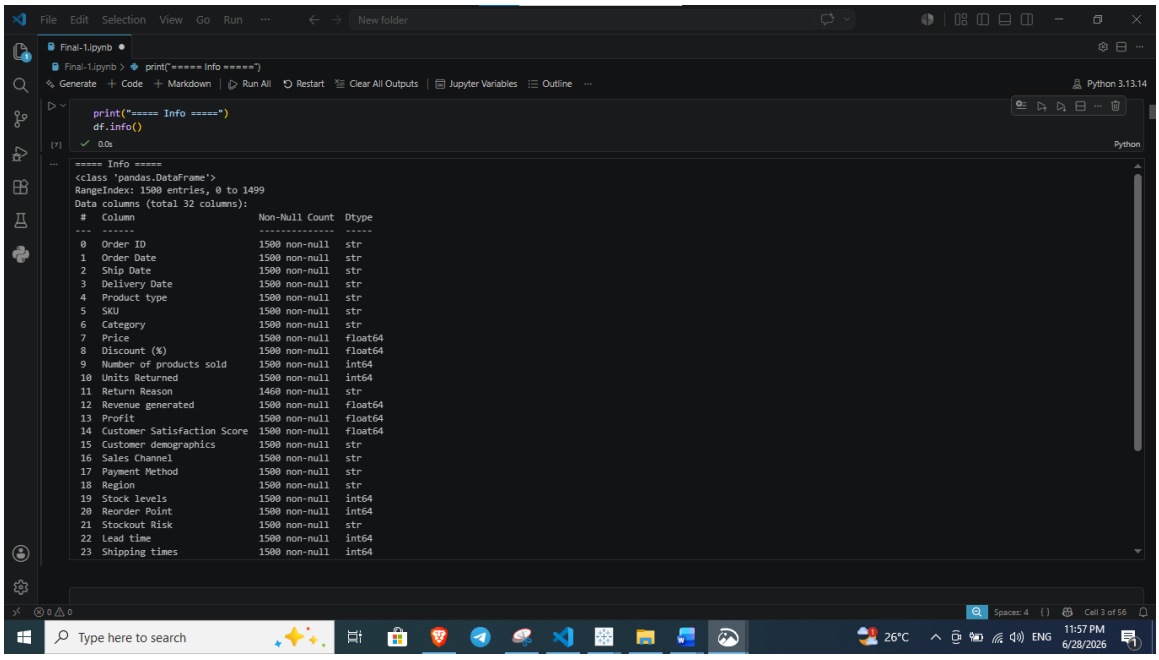
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5. Data Understanding (Initial Data Exploration)

5.1 Dataset Structure

Before applying any preprocessing techniques, an initial exploration of the dataset was conducted to understand its structure, assess data quality, and identify potential issues that could affect the analysis. This stage included examining the dataset structure, generating descriptive statistics, checking for missing values, evaluating feature uniqueness, and detecting duplicate records.

To better understand the dataset, its structure was inspected using the info() function. This step was performed to identify the number of records, available features, data types, and any missing values before starting the preprocessing phase.



The dataset contains **1,500 records** and **32 features**. The output also shows the data type of each feature, including categorical (object), numerical (int64 and float64), and date-related columns (which will be converted during preprocessing). Additionally, the Return Reason column contains **40 missing values**, indicating that some orders did not have a recorded return reason. This observation will be addressed during the data cleaning stage.

5.2 Descriptive Statistics

Descriptive statistics were generated using the describe() function to summarize the numerical features and understand their overall distribution before preprocessing.

```
print("===== Numerical Summary =====")
df.describe()
```

0.0s

Python

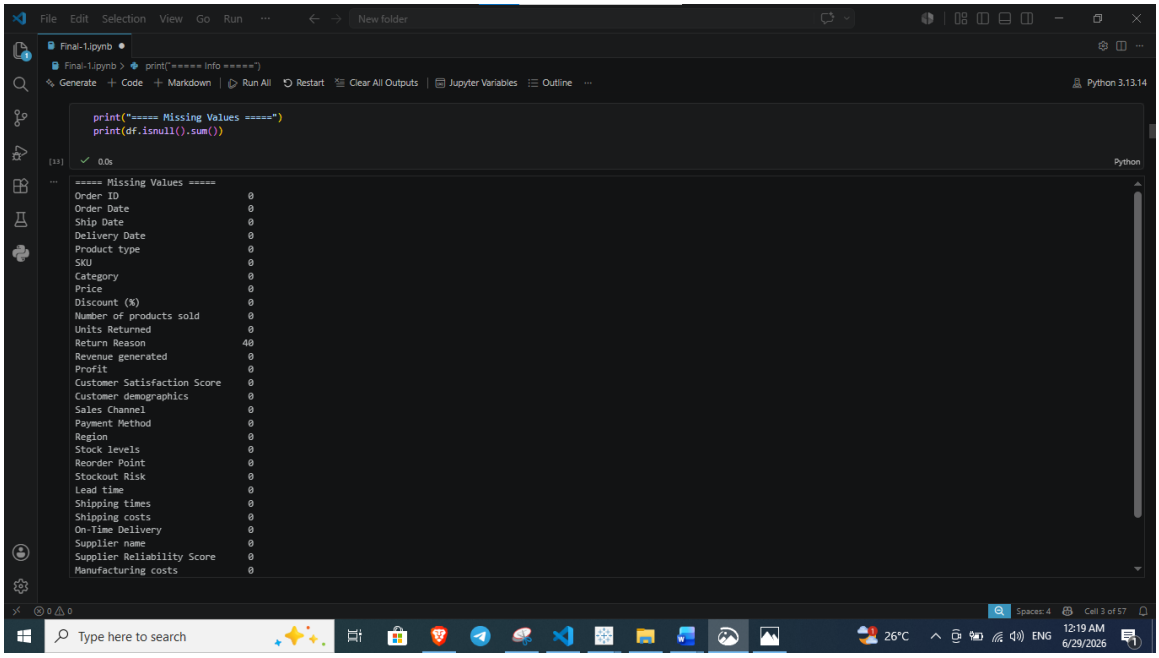
```
===== Numerical Summary =====
```

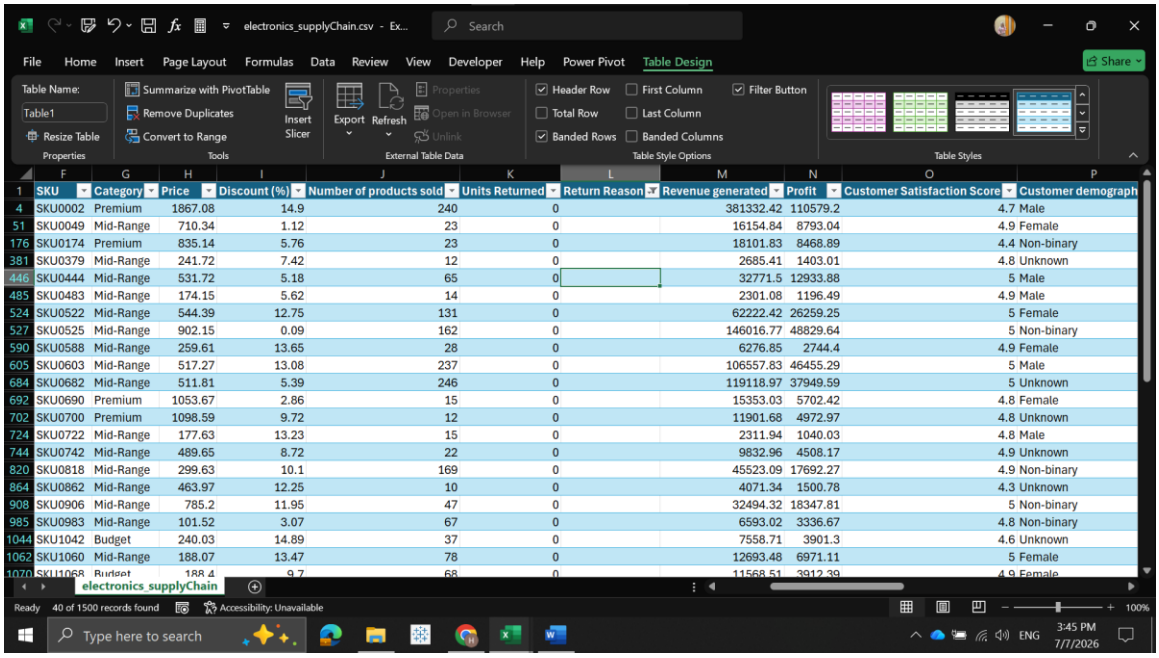
	Price	Discount (%)	Number of products sold	Units Returned	Revenue generated	Profit	Customer Satisfaction Score	Stock levels	Reorder Point	Lead time	Shipping times	Shipping costs	Supplier Reliability Score	Manufacturing costs	Quality Score
count	1500.000000	1500.000000	1500.000000	1500.000000	1.500000e+03	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000
mean	514.657507	7.549733	250.110000	18.373333	1.218708e+05	57084.533780	3.280133	153.940667	193.874000	13.488667	6.928667	176.448453	0.653045	221.836420	50.232133
std	474.288197	4.365707	142.960611	16.441437	1.542102e+05	72587.006805	1.032516	80.769464	139.756381	6.374608	3.731782	220.410392	0.144501	214.716303	28.905509
min	20.610000	0.010000	10.000000	0.000000	4.836800e+02	267.470000	1.200000	20.000000	3.000000	3.000000	1.000000	0.160000	0.310000	6.570000	0.110000
25%	172.275000	3.687500	126.000000	5.000000	2.700131e+04	12425.942500	2.400000	85.000000	81.000000	8.000000	4.000000	33.520000	0.550000	71.237500	25.072500
50%	370.555000	7.660000	246.000000	14.000000	6.584790e+04	30405.660000	3.300000	152.000000	166.000000	14.000000	7.000000	93.815000	0.648000	156.775000	50.175000
75%	707.325000	11.280000	375.000000	27.000000	1.508212e+05	72484.055000	4.200000	221.000000	279.000000	19.000000	10.000000	243.070000	0.757000	297.610000	76.272500
max	2499.950000	15.000000	498.000000	73.000000	1.178621e+06	648842.550000	5.000000	299.000000	694.000000	24.000000	13.000000	1589.800000	0.981000	1344.060000	99.900000

The statistical summary provides key descriptive measures such as count, mean, standard deviation, minimum, maximum, and quartiles for all numerical attributes. These statistics provide an initial understanding of the data distribution and help identify potential anomalies or outliers that may require further investigation during the preprocessing stage.

5.3 Missing Values Analysis

The dataset was examined for missing values to identify incomplete records and determine whether any data cleaning procedures would be required.





The results indicate that all features are complete except the **Return Reason** column, which contains **40 missing values**. This is expected because not all customer orders resulted in product returns. These missing values were handled during the data cleaning stage to ensure data consistency.

5.4 Duplicate Records Analysis

Duplicate records were checked to verify the integrity of the dataset and ensure that each order is represented only once.

```
print("Duplicates:", df.duplicated().sum())

print("Duplicates Order ID:", df['Order ID'].duplicated().sum())

display(df[df.duplicated()])

df = df.drop_duplicates()

print("After cleaning:", df.duplicated().sum())
```

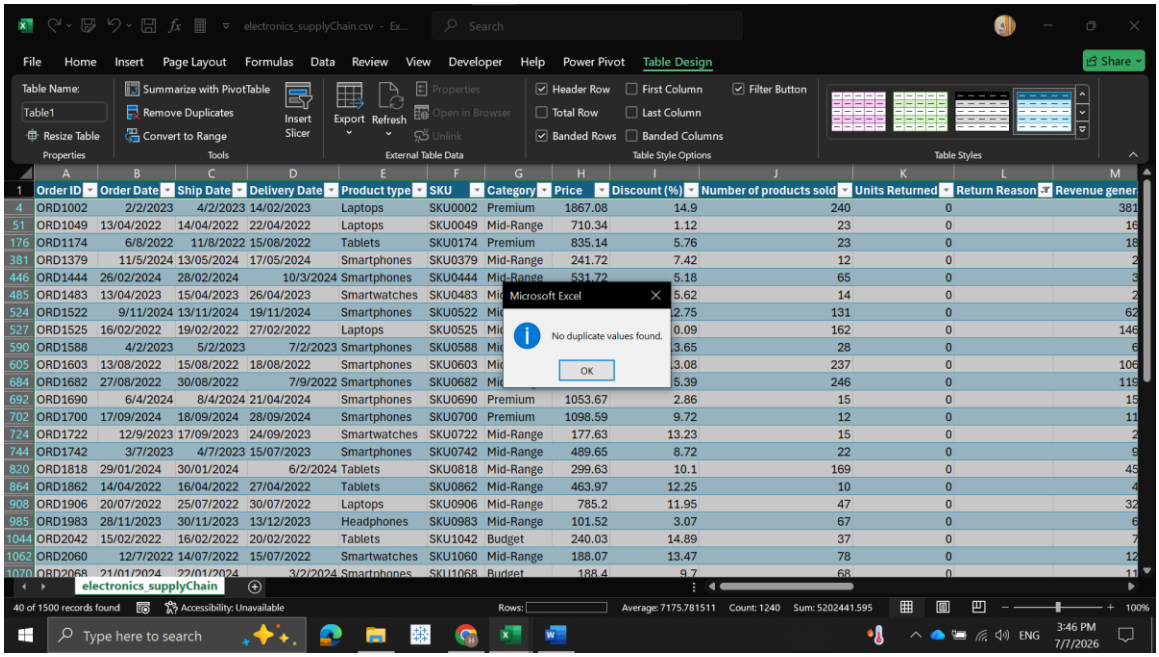
Python

```
Duplicates: 0
Duplicates Order ID: 0
```

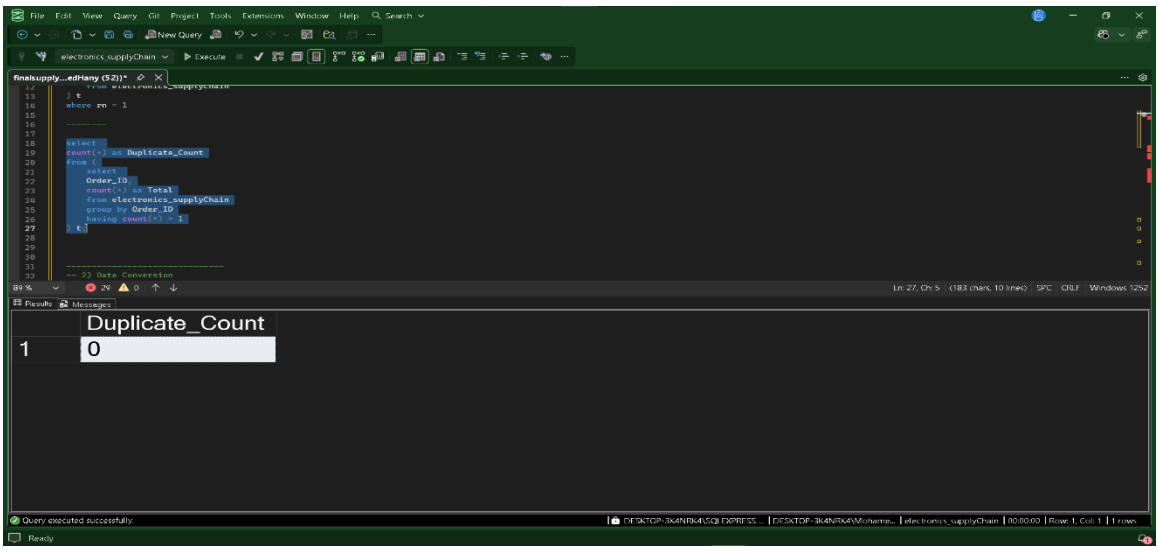
Order ID	Order Date	Ship Date	Delivery Date	Product type	SKU	Category	Price	Discount (%)	Number of products sold	...	Lead time	Shipping times	Shipping costs	On-Time Delivery	Supplier name	Supplier Reliability Score	Manufacturing costs	Quality Score	Defect rates	Production volumes

0 rows × 32 columns

```
After cleaning: 0
```



The duplicate analysis confirmed that the dataset contains **no duplicate records** and that all **Order IDs are unique**. As a result, no records were removed during this step, and the dataset maintained its original size.

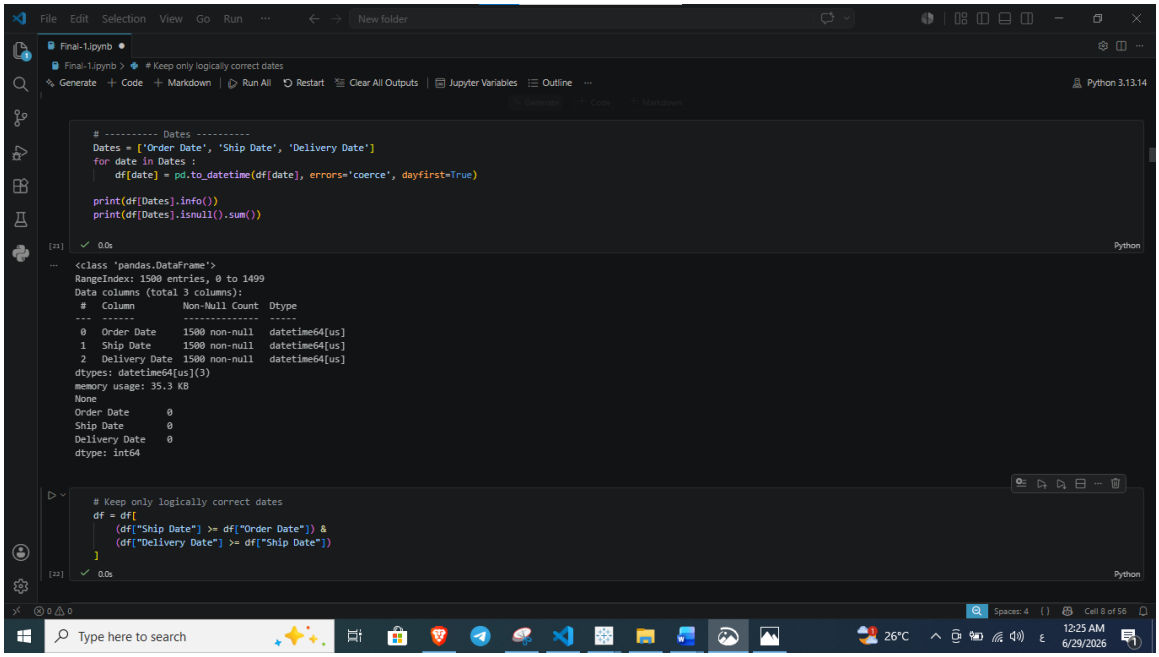


6. Data Cleaning & Preprocessing

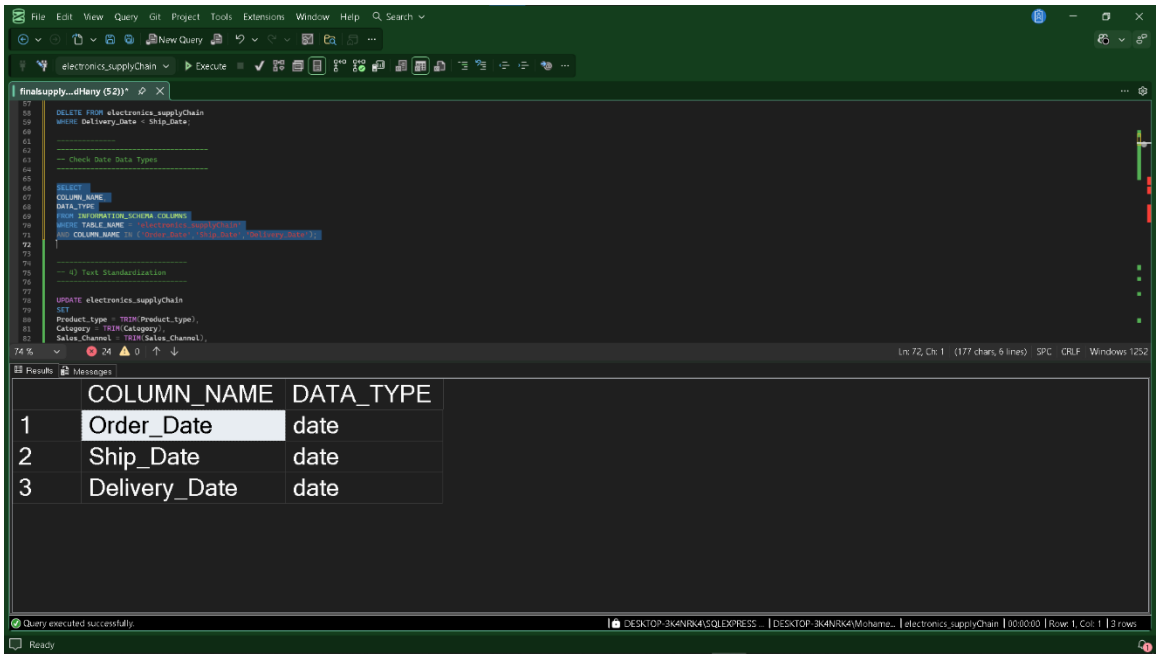
After exploring the dataset and identifying potential data quality issues, several preprocessing techniques were applied to improve data consistency, standardize the data format, and prepare the dataset for reliable analysis.

6.1 Date Processing and Validation

The date columns were converted into the datetime format to enable time-based analysis. Logical validation was also performed to ensure that each order followed the correct chronological sequence.

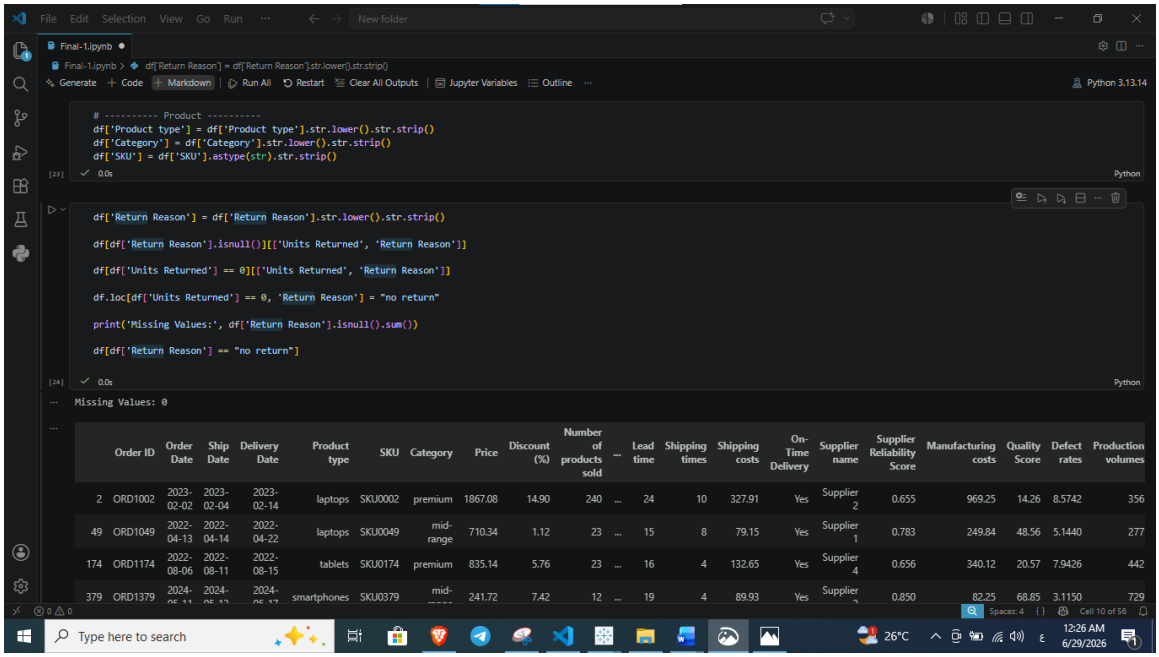


Order Date	Ship Date	Delivery Date
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
6/19/2022	6/23/2022	6/28/2022
3/3/2023	3/4/2023	3/7/2023
3/30/2022	3/31/2022	4/11/2022
2/2/2023	2/4/2023	2/14/2023
6/25/2024	6/30/2024	7/6/2024
2/28/2022	3/3/2022	3/14/2022
10/28/2022	10/31/2022	11/4/2022
5/20/2023	5/23/2023	5/29/2023
6/27/2024	7/2/2024	7/15/2024
10/8/2023	10/11/2023	10/23/2023
11/10/2024	11/12/2024	11/15/2024
6/17/2023	6/18/2023	6/19/2023
5/25/2023	5/30/2023	5/31/2023
1/11/2023	1/14/2023	1/15/2023
7/6/2023	7/8/2023	7/13/2023
4/22/2022	4/25/2022	5/7/2022
4/11/2022	4/12/2022	4/15/2022
1/21/2024	1/26/2024	1/27/2024



6.2 Data Standardization and Missing Value Handling

Categorical attributes were standardized to eliminate inconsistencies caused by different text formats. Missing values in the **Return Reason** column were also handled to maintain data completeness.



```
# ---------- Product ----------
df['Product type'] = df['Product type'].str.lower().str.strip()
df['Category'] = df['Category'].str.lower().str.strip()
df['SKU'] = df['SKU'].astype(str).str.strip()
```

```
df['Return Reason'] = df['Return Reason'].str.lower().str.strip()

df[df['Return Reason'].isnull()][['Units Returned', 'Return Reason']]

df[df['Units Returned'] == 0][['Units Returned', 'Return Reason']]

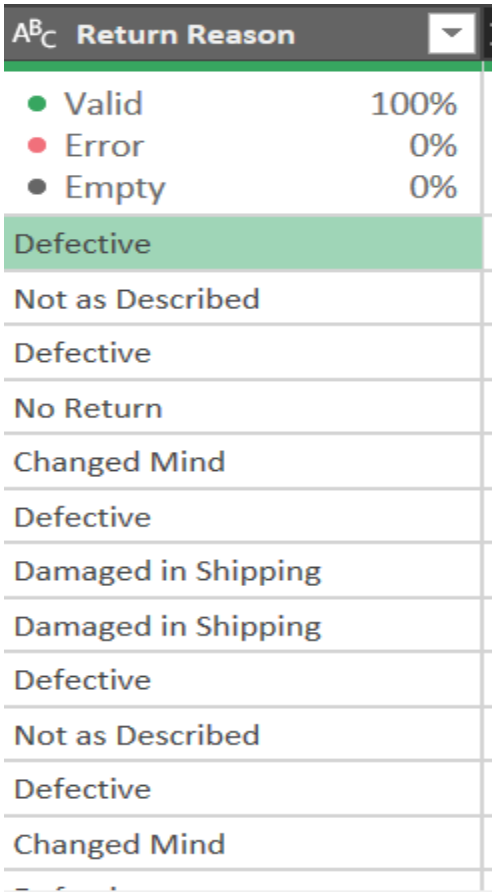
df.loc[df['Units Returned'] == 0, 'Return Reason'] = "no return"

print('Missing Values:', df['Return Reason'].isnull().sum())

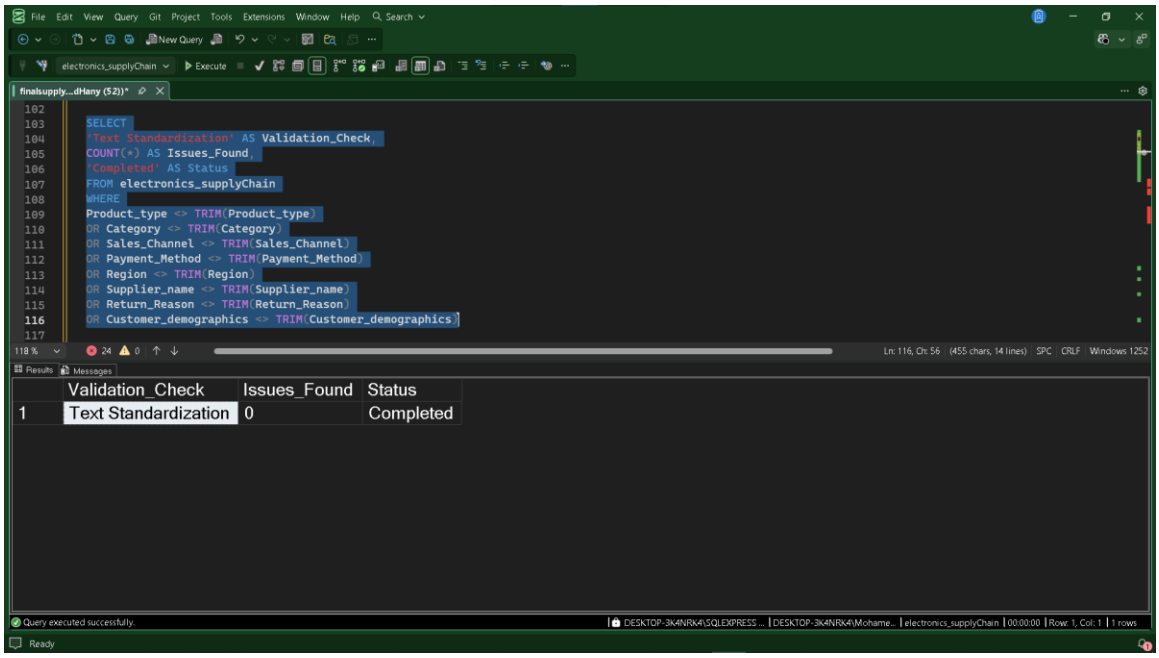
df[df['Return Reason'] == "no return"]
```

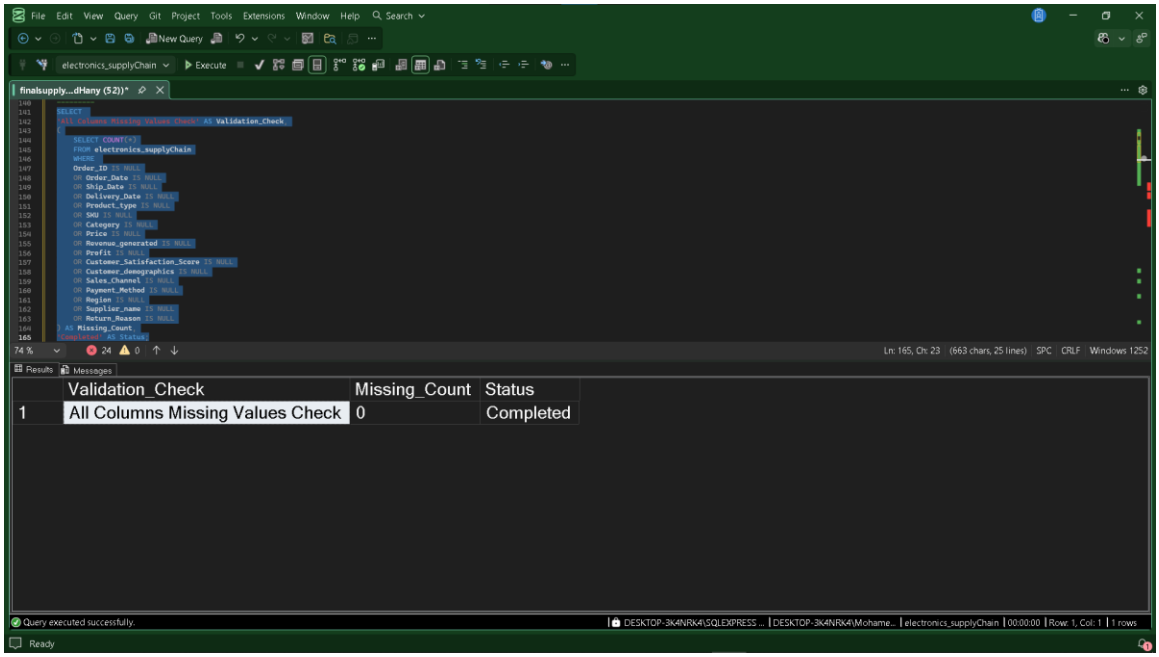
```
Missing Values: 0
```

	Order ID	Order Date	Ship Date	Delivery Date	Product type	SKU	Category	Price	Discount (%)	Number of products sold	...	Lead time	Shipping times	Shipping costs	On-Time Delivery	Supplier name	Supplier Reliability Score	Manufacturing costs	Quality Score	Defect rates	Production volumes
2	ORD1002	2023-02-02	2023-02-04	2023-02-14	laptops	SKU0002	premium	1867.08	14.90	240	...	24	10	327.91	Yes	Supplier 2	0.655	969.25	14.26	8.5742	356
49	ORD1049	2022-04-13	2022-04-14	2022-04-22	laptops	SKU0049	mid-range	710.34	1.12	23	...	15	8	79.15	Yes	Supplier 1	0.783	249.84	48.56	5.1440	277
174	ORD1174	2022-08-06	2022-08-11	2022-08-15	tablets	SKU0174	premium	835.14	5.76	23	...	16	4	132.65	Yes	Supplier 4	0.656	340.12	20.57	7.9426	442
379	ORD1379	2024-	2024-	2024-	smartphones	SKU0379	mid-	241.72	7.42	12	...	19	4	89.93	Yes	Supplier	0.850	82.25	68.85	3.1150	729



Return Reason	
Valid	100%
Error	0%
Empty	0%
Defective	
Not as Described	
Defective	
No Return	
Changed Mind	
Defective	
Damaged in Shipping	
Damaged in Shipping	
Defective	
Not as Described	
Defective	
Changed Mind	





Observation

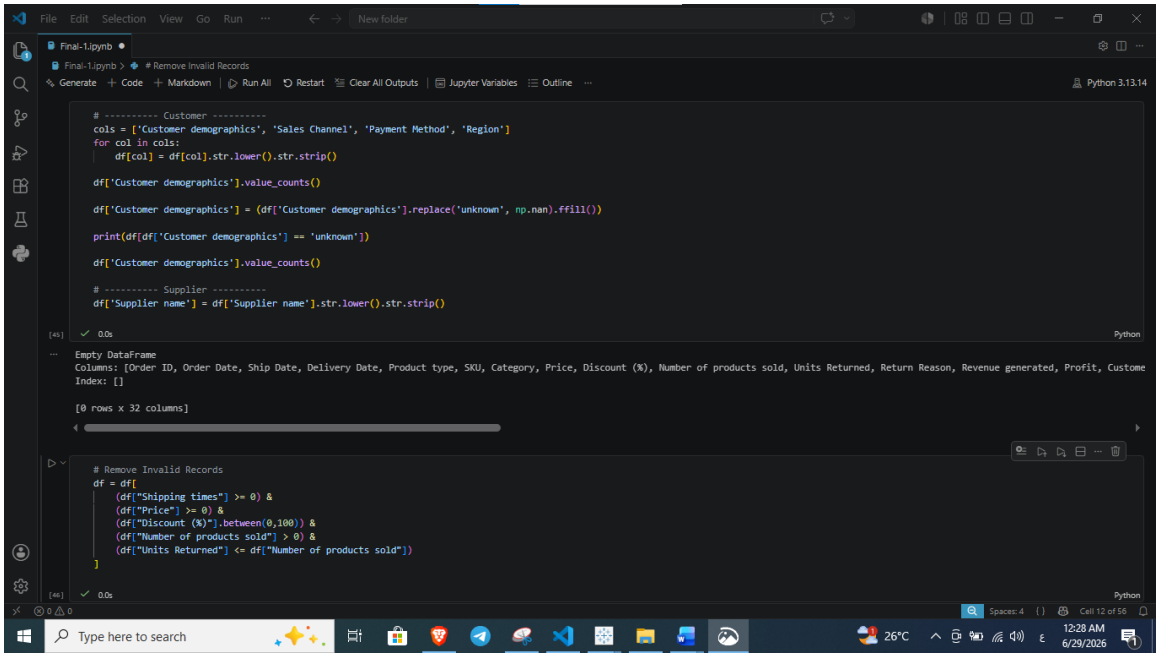
Text-based attributes were standardized by converting all values to lowercase and removing unnecessary spaces. Since orders with zero returned units do not require a return reason, missing values in the **Return Reason** column were replaced with **"no return"**, resulting in a complete and consistent categorical feature. Customer and supplier-related categorical fields were also standardized to improve data quality before analysis.

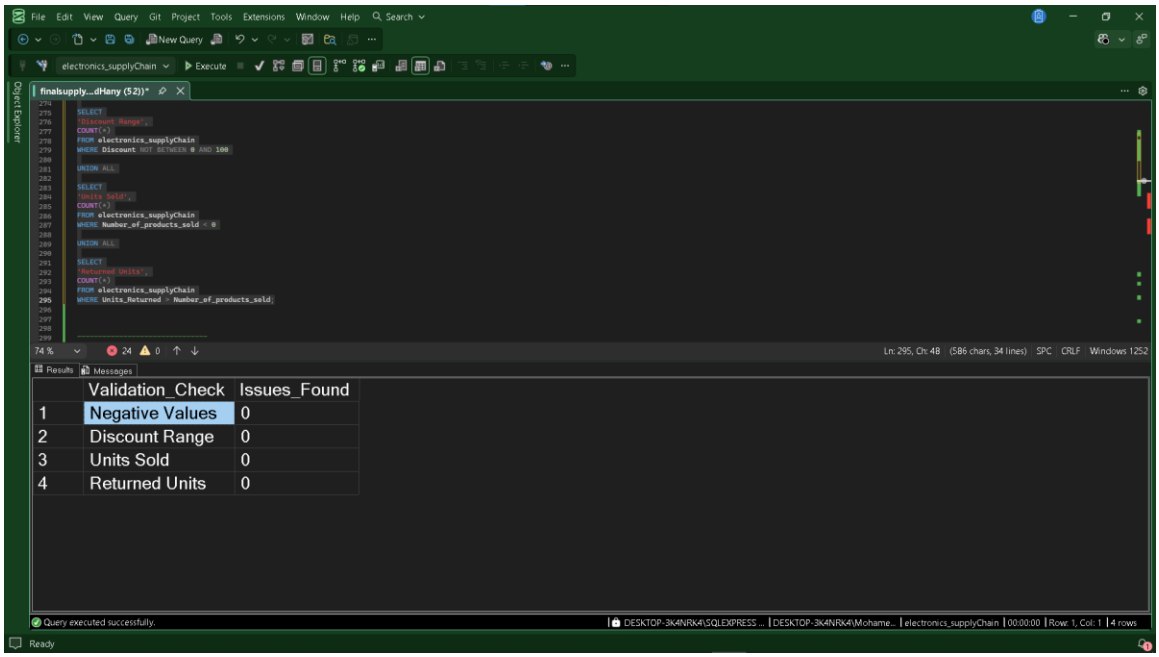
6.3 Removing Invalid Records

The dataset was filtered to remove records containing invalid or logically inconsistent values that could negatively affect the analytical results.

Observation

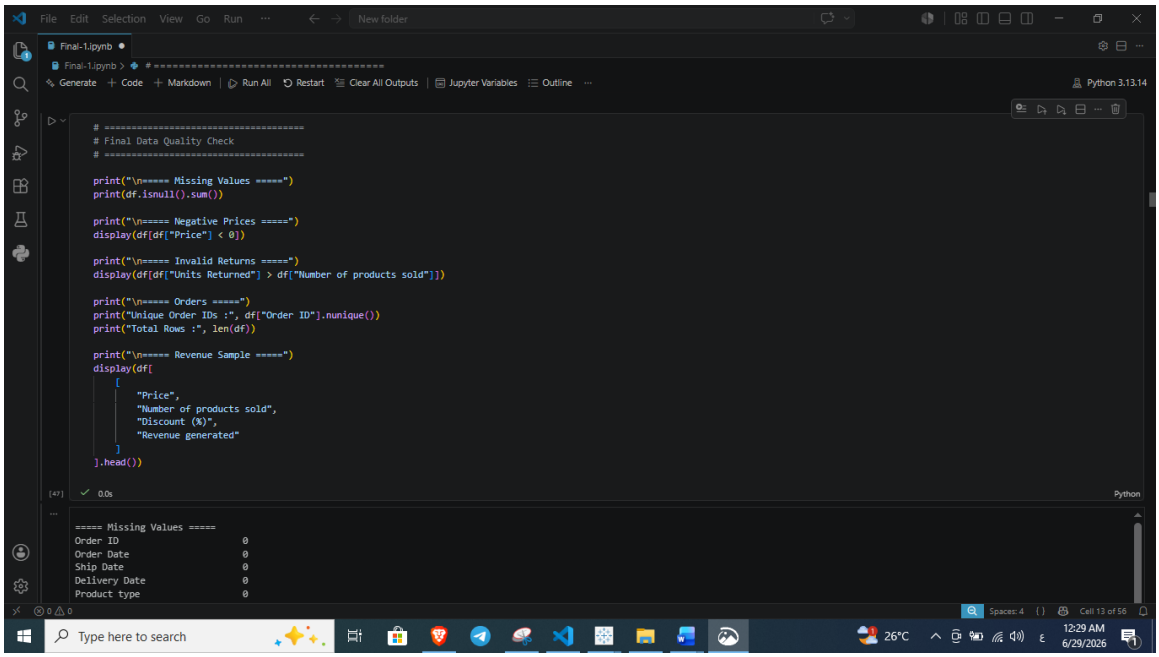
Several validation rules were applied to retain only valid observations. These rules ensured that shipping times and prices were non-negative, discount percentages fell within the valid range, the number of products sold was greater than zero, and returned units did not exceed the quantity sold. Applying these rules improved the overall reliability of the dataset.

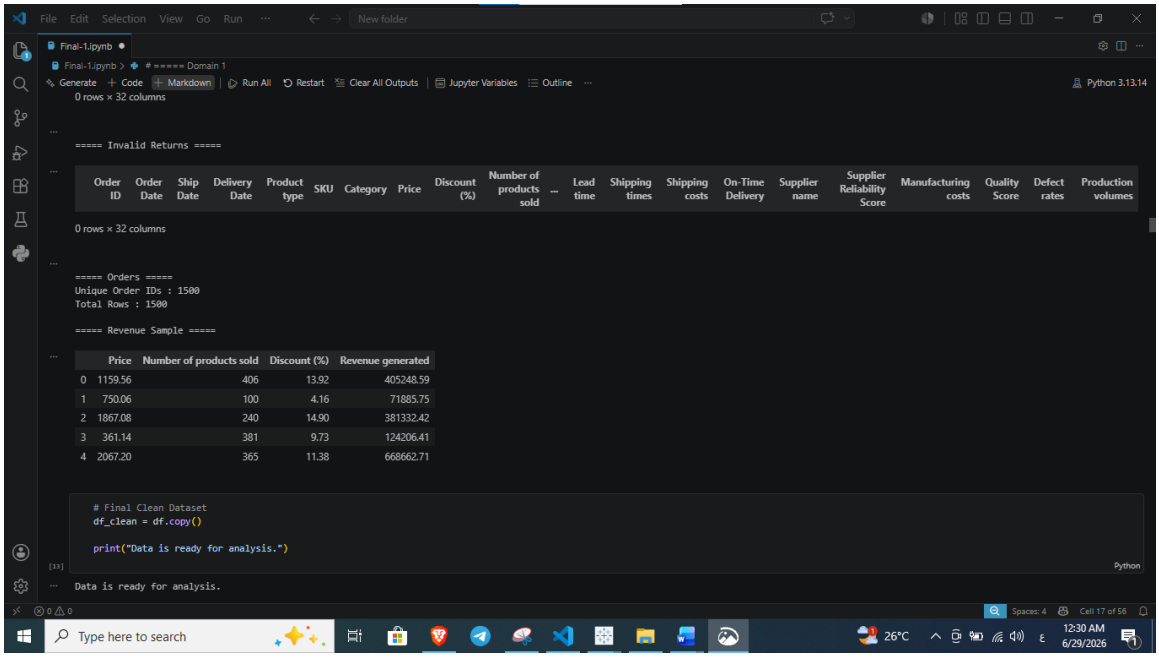




6.4 Final Data Validation

A final validation step was performed to verify the quality of the cleaned dataset before beginning the analytical phase.





Order ID		Order Date		Ship Date		Delivery Date		SKU		1.2	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%		
Error	0%	Error	0%	Error	0%	Error	0%	Error	0%		
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%		

1.2 Discount (%)		1²³ Number of products sold		1²³ Units Returned		Return Reason		1.2 Revenue generated	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%

1.2 Profit		1.2 Customer Satisfaction Score		Customer demographics		Sales Channel		Payment Me	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	
Error	0%	Error	0%	Error	0%	Error	0%	Error	
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	

Payment Method		Region		1²³ Lead time		1²³ Shipping times		1.2 Shipping costs	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%

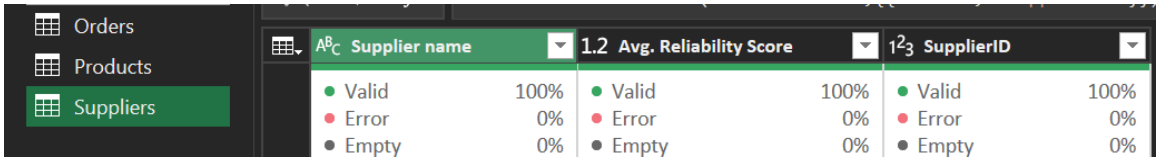
On-Time Delivery		Supplier name		1.2 Quality Score		1.2 Defect rates		1²³ SupplierID	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%

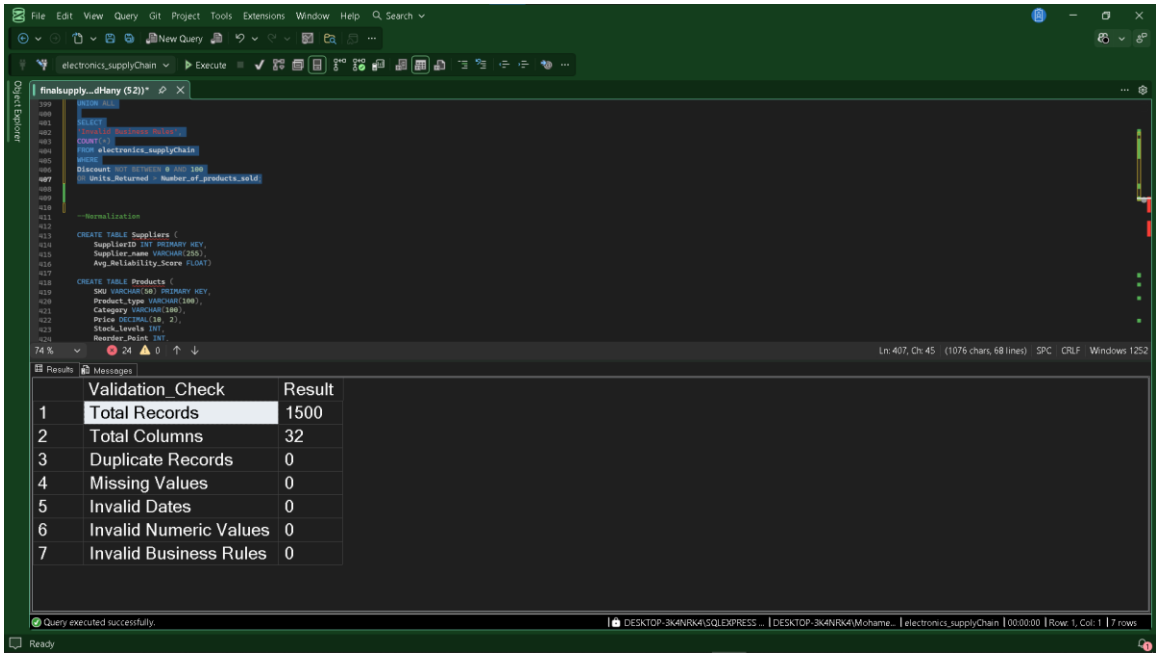
Products
Suppliers

SKU		Product type		Category		1.2 Price		1²³ Stock levels	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%

Products
Suppliers

1²³ Stock levels		1²³ Reorder Point		Stockout Risk		1.2 Manufacturing costs		1²³ Production volumes	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%





Observation

The final validation confirmed that the cleaned dataset contained no missing values in the required fields, no invalid prices, and no inconsistent return records. In addition, all order identifiers remained unique, and the cleaned dataset was copied into a new DataFrame (df_clean) to preserve a validated version for the exploratory data analysis stage.

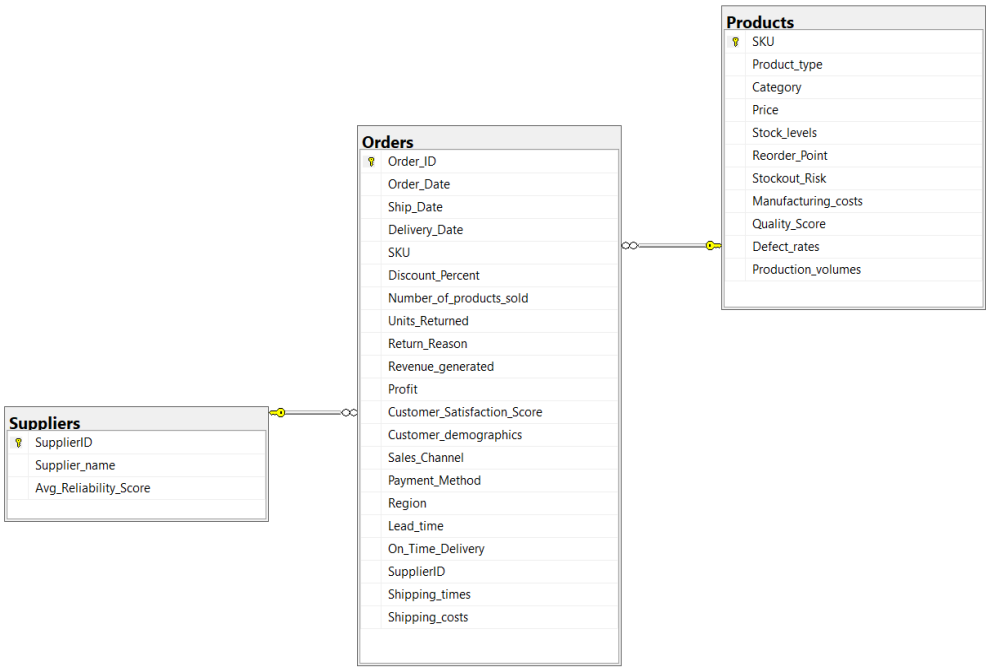
Summary of Data Preprocessing

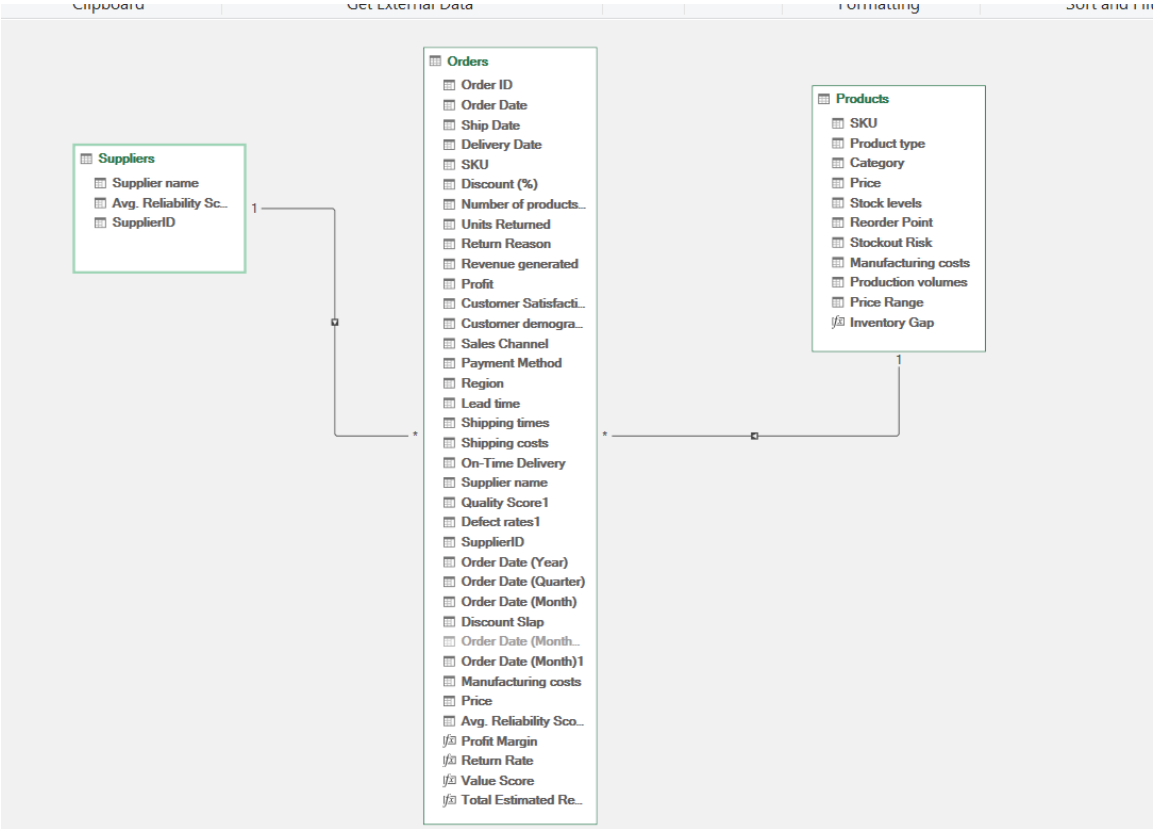
Step	Status
Duplicate Records Removed	✅ Completed
Date Conversion	✅ Completed
Date Validation	✅ Completed
Text Standardization	✅ Completed

Missing Value Handling	✅ Completed
Invalid Records Removal	✅ Completed
Final Data Validation	✅ Completed
Analysis Dataset Created (df_clean)	✅ Completed

Data Normalization

Following the initial data cleaning and validation phases, Data Normalization was implemented as a critical step to ensure analytical integrity and equitable comparison. This process aims to rescale numeric data into a unified range, preventing variables with larger magnitudes from disproportionately influencing the analytical models. This is particularly essential when integrating diverse performance indicators such as manufacturing costs, quality scores, and defect rates found within the database schema





The schema highlights the structured relationship between the Suppliers, Orders, and Products entities. The normalization process applied to this dataset achieved the following:

Standardization of Metrics: By rescaling numeric attributes across the tables—specifically within the Products and Suppliers entities—we ensured that variables with different units of measurement are comparable.

Enhanced Analytical Precision: Normalization mitigated the impact of scale disparities (e.g., between Manufacturing_costs and Quality_Score), allowing for a more accurate evaluation of supplier performance.

Foundation for EDA: This step served as the essential prerequisite for the Exploratory Data Analysis (EDA) stage, ensuring that the aggregated data used to derive Key Performance Indicators (KPIs) remains robust, reliable, and ready for advanced business insights.

7. Exploratory Data Analysis (EDA)

Overview

After completing the data cleaning and validation processes, the cleaned dataset (df_clean) was used to perform Exploratory Data Analysis (EDA). The objective of this stage was to summarize the data, calculate key performance indicators (KPIs), and answer business-related questions through statistical aggregation and descriptive analysis.

The analysis was organized into multiple business domains, with each domain focusing on a specific operational area of the supply chain.

The domains included:

- Commercial Insights
- Supplier Performance Analysis
- Customer Experience Analysis
- Inventory Analysis
- Shipping & Delivery Analysis

---

8. Domain 1: Commercial Insights

Overview

The Commercial Insights analysis focuses on evaluating the overall financial performance of the business. It examines revenue, profitability, product performance, sales channels, regional performance, customer demographics, and discount behavior to provide a comprehensive understanding of the company's commercial operations.

Business Questions

The following business questions were addressed during this analysis:

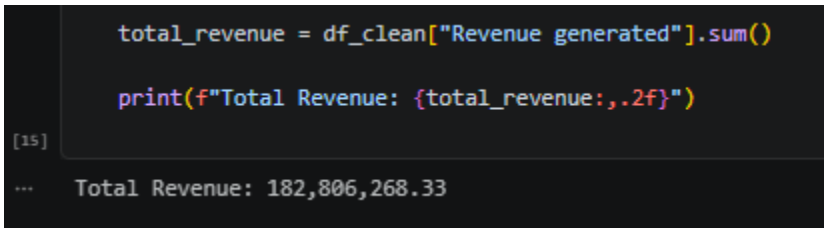
1. What is the total revenue?
2. What is the total profit?
3. What is the profit margin?
4. What is the average discount percentage?
5. How have revenue and profit changed quarterly?

6. Which product generates the highest revenue and profit?
7. Which sales channel generates the highest profit?
8. Which region has the highest profit margin?
9. How are discounts distributed?
10. How do revenue and profit vary by customer demographics?
11. How do revenue and profit vary across discount categories?

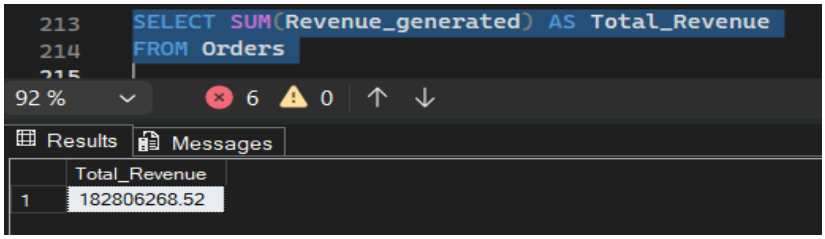
8.1 Total Revenue

The total revenue was calculated by summing all values in the **Revenue generated** column to determine the overall sales generated during the analysis period.

Python Calculation



SQL Validation



Excel:

Sum of Revenue generated
$182,806,268.33

The analysis shows that the business generated a total revenue of **182,806,268.33**, representing the total sales value across all customer orders.

8.2 Total Profit

The total profit was calculated by summing all values in the **Profit** column to measure the overall profitability of the business.

Python Calculation

```
total_profit = df_clean["Profit"].sum()

print(f"Total Profit: {total_profit:,.2f}")
```

[16]

```
Total Profit: 85,626,800.67
```

SQL Validation

```
215
216    SELECT SUM(Profit) AS Total_Profit
217    FROM Orders
218
```

92 % 24 0

Results Messages

	Total_Profit
1	85626800.60

Excel

Sum of Profit
$85,626,800.67

The analysis indicates a total profit of **85,626,800.67**, representing the net earnings generated from all completed orders.

8.3 Profit Margin

The profit margin was calculated as the ratio between total profit and total revenue to evaluate the overall profitability of the business.

Python Calculation

```
profit_margin = (
    total_profit / total_revenue
) * 100

print(f"Profit Margin: {profit_margin:.2f}%")
```

[17]

```
Profit Margin: 46.84%
```

SQL Validation

```
224   SELECT
225   AVG( [Discount_Percent] ) AS Average_Discount
226   FROM Orders
227
```

92 % 24 0

Results Messages

	Profit_Margin
1	46.840188

Excel

Table name: Orders

Measure name: Profit Margin

Description:

Formula: fx Check formula

=SUM(Orders[Profit])/SUM(Orders[Revenue generated])

SUM(**ColumnName**)

Profit Margin
46.84%
46.84%

8.4 Average Discount Percentage

The average discount percentage was calculated to determine the typical discount offered across all customer orders. This KPI provides an overview of the company's pricing and promotional strategy.

Python Calculation

```
    avg_discount = df_clean["Discount (%)"].mean()

    print(f"Average Discount: {avg_discount:.2f}%")
18]
...  Average Discount: 7.55%
```

SQL Validation

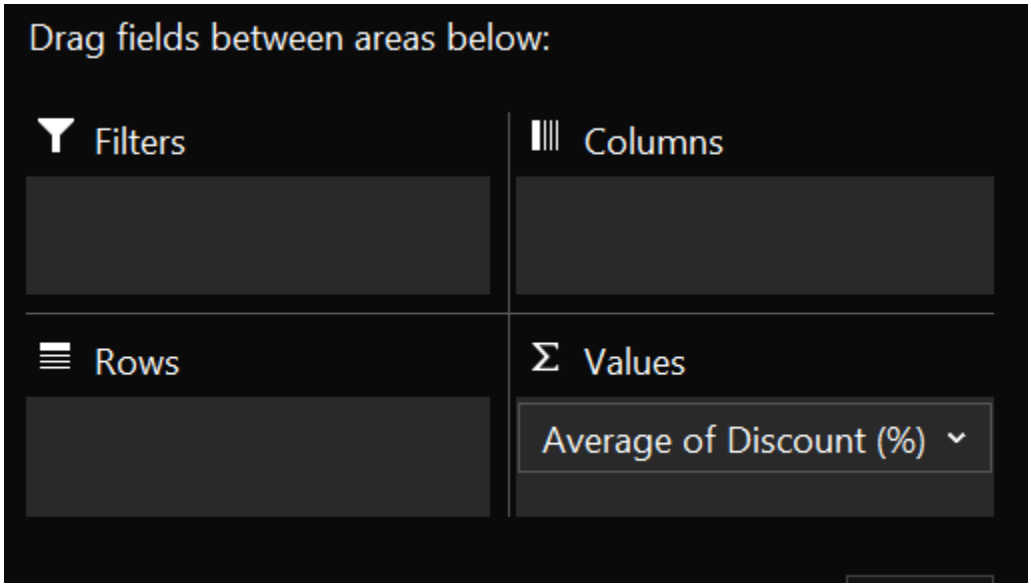
```
222    SELECT
223    AVG( [Discount_Percent] ) AS Average_Discount
224    FROM Orders
```

92 % 24 0

Results Messages

	Average_Discount
1	7.549733

Excel



Average of Discount (%)
7.55

The analysis shows that the average discount percentage is **7.55%**, indicating that customers received an average discount of approximately **7.5%** per order.

8.5 Revenue and Profit by Quarter

Quarterly revenue and profit were calculated by extracting the quarter from the order date and aggregating both revenue and profit for each quarter. This analysis helps evaluate business performance trends over time.

Python Calculation

```python
df_clean["Quarter"] = df_clean["Order Date"].dt.to_period("Q")

quarterly_analysis = (
    df_clean
    .groupby("Quarter")[["Revenue generated", "Profit"]]
    .sum()
    .reset_index()
)

quarterly_analysis
```

SQL Validation

```sql
-- Quarterly Trend
WITH QuarterlyData AS (
    SELECT
       DATEPART(QUARTER, Order_Date) AS Quarter_Number,
        'Q' + CAST(DATEPART(QUARTER, Order_Date) AS VARCHAR(1)) AS Quarter_Name,
        SUM(Revenue_generated) AS Total_Revenue,
        SUM(Profit) AS Total_Profit
    FROM Orders
    GROUP BY DATEPART(QUARTER, Order_Date)
)
SELECT
    Quarter_Name,
    Total_Revenue,
    Total_Profit
FROM QuarterlyData
ORDER BY Quarter_Number
```

92 % 24 0

Results Messages

	Quarter_Name	Total_Revenue	Total_Profit
1	Q1	40947473.06	19330932.75
2	Q2	44912454.59	20687838.65
3	Q3	45162679.04	21449982.30
4	Q4	51783661.83	24158046.90

Excel

Row Labels	Sum of Revenue generated	Sum of Profit
2022		
Qtr1	$ 14,580,525.56	$ 6,710,772.14
Qtr2	$ 16,403,959.63	$ 7,573,410.86
Qtr3	$ 14,237,676.70	$ 6,911,348.93
Qtr4	$ 14,935,634.88	$ 7,049,973.44
2023		
Qtr1	$ 14,785,271.48	$ 7,088,232.32
Qtr2	$ 11,679,194.83	$ 5,362,412.86
Qtr3	$ 16,918,444.26	$ 7,805,106.84
Qtr4	$ 17,297,217.06	$ 7,968,683.48
2024		
Qtr1	$ 11,581,675.95	$ 5,531,928.25
Qtr2	$ 16,829,300.08	$ 7,752,014.92
Qtr3	$ 14,006,558.08	$ 6,733,526.60
Qtr4	$ 19,550,809.82	$ 9,139,390.03
Grand Total	**$ 182,806,268.33**	**$ 85,626,800.67**

The quarterly analysis summarizes the total revenue and profit generated during each quarter from **2022 Q1** through **2024 Q4**. The results show fluctuations in financial performance across different quarters, enabling comparisons of seasonal trends and identifying periods of stronger or weaker business performance.

8.6 Revenue and Profit by Product

Revenue and profit were aggregated by product type to identify the products that contribute the most to the company's financial performance. The results were then sorted in descending order based on revenue and profit.

Python Calculation

```python
product_analysis = (
    df_clean
    .groupby("Product type")[["Revenue generated", "Profit"]]
    .sum()
    .sort_values(
        by=["Revenue generated", "Profit"],
        ascending=False
    )
)

product_analysis
```

SQL Validation

```sql
-- Revenue and profitability per product
SELECT
    p.Product_type,
    SUM(o.Revenue_generated) AS Total_Revenue,
    SUM(o.Profit) AS Total_Profit,
    (SUM(o.Profit) * 1.0 / NULLIF(SUM(o.Revenue_generated),0)) * 100 AS Profit_Margin_Percent
FROM Orders o
JOIN Products p ON o.SKU = p.SKU
GROUP BY p.Product_type
ORDER BY Total_Revenue DESC
```

92 % 24 0

Results Messages

	Product_type	Total_Revenue	Total_Profit	Profit_Margin_Percent
1	laptops	79662632.02	37114017.20	46.588900
2	smartphones	40783571.57	19529732.98	47.886200
3	tablets	34638181.37	15929751.48	45.988900
4	smartwatches	17205532.33	8145736.06	47.343700
5	headphones	10516351.23	4907562.88	46.666000

Excel

Row Labels	Sum of Revenue generated	Sum of Profit
Headphones	$ 10,516,351.23	$ 4,907,562.88
Smartwatches	$ 17,205,532.26	$ 8,145,736.06
Tablets	$ 34,638,181.20	$ 15,929,751.48
Smartphones	$ 40,783,571.57	$ 19,529,732.95
Laptops	$ 79,662,632.07	$ 37,114,017.30
Grand Total	**$ 182,806,268.33**	**$ 85,626,800.67**

The analysis shows that **Laptops** generated the highest revenue and profit among all product categories, followed by **Smartphones** and **Tablets**. In contrast, **Headphones**

contributed the lowest revenue and profit, indicating a relatively smaller financial impact compared to the other product types.

8.7 Profit by Sales Channel

Profit and the total number of products sold were aggregated for each sales channel to evaluate channel performance and identify the most profitable sales source.

Python Calculation

```python
sales_channel_profit = (
    df_clean
    .groupby("Sales Channel")[["Profit", "Number of products sold"]]
    .sum()
    .sort_values(by="Profit", ascending=False)
    .reset_index()
)

sales_channel_profit
```

SQL Validation

```sql
-- Sales Channel Revenues and Profitability
SELECT
    Sales_Channel,
    SUM(Revenue_generated) AS Total_Revenue,
    SUM(Profit) AS Total_Profit,
    SUM(Number_of_products_sold) AS Total_Volume
FROM Orders
GROUP BY Sales_Channel
```

92 % 24 0

Results Messages

	Sales_Channel	Total_Revenue	Total_Profit	Total_Volume
1	online	38438408.28	18265991.12	75250
2	b2b	37237898.25	16882203.30	80519
3	wholesale	36163778.79	17135613.90	70208
4	direct sales	35368044.86	16251009.19	76018
5	retail store	35598138.34	17091983.09	73170

Excel

Row Labels	Sum of Profit	Sum of Number of products sold
B2B	$ 16,882,203.33	$ 80,519.00
Direct Sales	$ 16,251,009.18	$ 76,018.00
Online	$ 18,265,991.14	$ 75,250.00
Retail Store	$ 17,091,983.08	$ 73,170.00
Wholesale	$ 17,135,613.94	$ 70,208.00
Grand Total	**$ 85,626,800.67**	**$ 375,165.00**

The analysis indicates that the **Online** sales channel generated the highest total profit, followed by **Wholesale**, **Retail Store**, **B2B**, and **Direct Sales**. This comparison highlights the relative contribution of each sales channel to the company's overall profitability while also showing the corresponding sales volume.

8.8 Profit Margin by Region

The profit margin was calculated for each region by dividing the total profit by the total revenue. This analysis helps compare regional profitability and identify the regions generating the highest financial returns.

Python Calculation

```
region_profit_margin = (
    df_clean
    .groupby("Region")
    .agg({
        "Revenue generated":"sum",
        "Profit":"sum"
    })
)

region_profit_margin["Profit Margin (%)"] = (
    region_profit_margin["Profit"] /
    region_profit_margin["Revenue generated"]
) * 100

region_profit_margin = (
    region_profit_margin
    .sort_values(
        by="Profit Margin (%)",
        ascending=False
    )
)

region_profit_margin["Profit Margin (%)"]
```

SQL Validation

```sql
-- Profit margin per region
SELECT
    Region,
    SUM(Profit) AS Total_Profit,
    SUM(Revenue_generated) AS Total_Revenue,
    (SUM(Profit) * 1.0 / NULLIF(SUM(Revenue_generated),0)) * 100 AS Profit_Margin_Percent
FROM Orders
GROUP BY Region
ORDER BY Total_Profit DESC
```

92 % 24 0

Results Messages

	Region	Total_Profit	Total_Revenue	Profit_Margin_Percent
1	north	18268205.79	38859117.10	47.011300
2	west	17792653.25	37391468.00	47.584700
3	south	17341509.83	35563459.91	48.762100
4	central	16288015.17	35992861.77	45.253400
5	east	15936416.56	34999361.74	45.533400

Excel

Row Labels	Profit Margin
Central	45.3%
East	45.5%
North	47.0%
South	48.8%
West	47.6%
Grand Total	**46.8%**

The regional analysis shows that the **South** region achieved the highest profit margin (**48.76%**), followed by the **West** (**47.58%**) and **North** (**47.01%**). The **East** and **Central** regions recorded relatively lower profit margins, indicating differences in regional profitability across the business.

8.9 Discount Distribution

Discounts were grouped into three categories to analyze their distribution across customer orders.

Python Calculation

```python
# ====================================
# Q9: Discount Distribution (%)
# ====================================

df_clean["Discount Category"] = pd.cut(
    df_clean["Discount (%)"],
    bins=[0, 5, 10, 15],
    labels=["Low", "Medium", "High"],
    include_lowest=True
)

discount_distribution = (
    df_clean["Discount Category"]
    .value_counts(normalize=True)
    .sort_index()
    .mul(100)
    .round(2)
    .reset_index()
)

discount_distribution.columns = [
    "Discount Category",
    "Percentage (%)"
]

discount_distribution
```

	Discount Category	Percentage (%)
0	Low	33.2
1	Medium	31.4
2	High	35.4

SQL Validation

```sql
-- % Discount on products sold
WITH DiscountRanges AS (
    SELECT
        CASE
            WHEN Discount_Percent BETWEEN 0 AND 5 THEN 'Low Discount (0-5%)'
            WHEN Discount_Percent BETWEEN 5 AND 10 THEN 'Medium Discount (5-10%)'
            ELSE 'High Discount (Above 10%)'
        END AS Discount_Category,
        Number_of_products_sold
    FROM Orders
),
CategoryTotals AS (
    SELECT
        Discount_Category,
        SUM(Number_of_products_sold) AS Category_Volume
    FROM DiscountRanges
    GROUP BY Discount_Category
),
GrandTotal AS (
    SELECT SUM(Number_of_products_sold) AS Total_Volume
    FROM Orders
)
SELECT
    ct.Discount_Category,
    ct.Category_Volume,
    CAST(100.0 * ct.Category_Volume / gt.Total_Volume AS DECIMAL(10,2)) AS Percentage_Of_Total
FROM CategoryTotals ct
CROSS JOIN GrandTotal gt
ORDER BY Percentage_Of_Total DESC
```

92 % ⌄ ⊗ 15 ⚠ 0 ↑ ↓

Results | Messages

	Discount_Category	Category_Volume	Percentage_Of_Total
1	High Discount (Above 10%)	129486	34.51
2	Low Discount (0-5%)	126705	33.77
3	Medium Discount (5-10%)	118974	31.71

Excel

Row Labels	Sum of Number of products sold
High	34.5%
Low	33.8%
Medium	31.7%
Grand Total	**100.0%**

The **High** discount category represents the largest share of orders (**35.4%**), followed by **Low** (**33.2%**) and **Medium** (**31.4%**), indicating a relatively balanced discount distribution.

8.10 Revenue and Profit by Customer Demographics

Revenue and profit were analyzed across customer demographic groups to compare their financial contribution.

Python Calculation

```python
gender_analysis = (
    df_clean
    .groupby("Customer demographics")[["Revenue generated","Profit"]]
    .sum()
    .sort_values(
        by="Revenue generated",
        ascending=False
    )
)

gender_analysis
```

SQL Validation

```sql
-- Segments Revenue and Profit
SELECT
    Customer_demographics,
    SUM(Revenue_generated) AS Total_Revenue,
    SUM(Profit) AS Total_Profit
FROM Orders
GROUP BY Customer_demographics
ORDER BY Total_Revenue DESC
```

92 % 23 0

Results Messages

	Customer_demographics	Total_Revenue	Total_Profit
1	male	63224894.02	29275399.87
2	female	61668230.71	29453804.33
3	non-binary	57913143.79	26897596.40

Excel

Row Labels	Sum of Revenue generated	Sum of Profit
Female	$ 61,668,230.62	$ 29,453,804.33
Male	$ 63,224,893.93	$ 29,275,399.97
Non-binary	$ 57,913,143.78	$ 26,897,596.37
Grand Total	**$ 182,806,268.33**	**$ 85,626,800.67**

The **Male** customer segment generated the highest revenue and profit, followed by **Female**, while the **Non-binary** segment recorded the lowest financial contribution.

8.11 Revenue and Profit by Discount Category

Revenue and profit were compared across the predefined discount categories to evaluate the financial impact of different discount levels.

Python Calculation

```python
discount_analysis = (
    df_clean
    .groupby("Discount Category")[["Revenue generated", "Profit"]]
    .sum()
    .reset_index()
)
```

SQL Validation

```sql
-- Revenue and Profit by Discount Level
WITH DiscountRanges AS (
    SELECT
        CASE
            WHEN Discount_Percent BETWEEN 0 AND 5 THEN 'Low Discount (0-5%)'
            WHEN Discount_Percent BETWEEN 5 AND 10 THEN 'Medium Discount (5-10%)'
            ELSE 'High Discount (Above 10%)'
        END AS Discount_Category,
        Revenue_generated,
        Profit
    FROM Orders
)
SELECT
    Discount_Category,
    SUM(Revenue_generated) AS Total_Revenue,
    SUM(Profit) AS Total_Profit
FROM DiscountRanges
GROUP BY Discount_Category
HAVING SUM(Profit) > 0
```

92 % 23 0

Results Messages

	Discount_Category	Total_Revenue	Total_Profit
1	Medium Discount (5-10%)	56632032.10	26545040.52
2	Low Discount (0-5%)	65675940.46	32376267.61
3	High Discount (Above 10%)	60498295.96	26705492.47

Excel

Row Labels	Sum of Revenue generated	Sum of Profit
High	$ 60,498,296.02	$ 26,705,492.44
Low	$ 65,675,940.40	$ 32,376,267.65
Medium	$ 56,632,031.91	$ 26,545,040.58
Grand Total	**$ 182,806,268.33**	**$ 85,626,800.67**

The **Low Discount** category generated the highest revenue and profit, followed by the **High Discount** category, while the **Medium Discount** category recorded the lowest overall financial performance.

8.12 Domain Summary

The Commercial Insights analysis evaluated the organization's overall sales performance by examining key financial metrics such as revenue, profit, profit margin, and average discount. The analysis also explored business performance across different quarters, product types, sales channels, regions, customer demographics, and discount categories.

Overall, the analysis provides a comprehensive view of business performance and highlights the main factors influencing revenue generation and profitability.

Key Insights

- The business generated a total revenue of **$182.81M** with a total profit of **$85.63M**.
- The overall profit margin reached **46.84%**, indicating strong profitability.
- The average discount offered across all orders was **7.55%**.
- Revenue and profit showed noticeable variations across different quarters, reflecting seasonal sales trends.
- **Laptops** generated the highest revenue and profit among all product categories.
- The **Online** sales channel achieved the highest total profit.
- The **South** region recorded the highest profit margin.
- Discount levels were relatively balanced across the three discount categories.
- Revenue and profit were distributed fairly evenly across customer demographic groups.
- Orders with **Low Discounts (0–5%)** generated the highest revenue and profit, indicating stronger profitability with lower discount levels.

Business Value

This analysis provides a clear understanding of the company's commercial performance by identifying the primary drivers of revenue and profitability. The findings support strategic decisions related to pricing, discount policies, product portfolio optimization,

regional sales performance, and sales channel effectiveness, ultimately helping improve business profitability and growth.

9. Domain 2: Customer Experience & Returns

9.1 Overview

This domain focuses on evaluating customer experience and product return behavior. The analysis examines customer satisfaction, return rates, return losses, sales channels, and return reasons to better understand the factors influencing product returns and customer experience. The generated insights help identify areas for improving customer satisfaction and reducing return-related costs.

Business Questions

The following business questions were addressed during this analysis:

1. What is the average return rate?
2. How many units were returned?
3. What is the average customer satisfaction score?
4. What is the total estimated return loss?
5. How does customer satisfaction vary by customer demographics?
6. Which region has the highest return rate and return loss?
7. Which sales channel has the highest return rate and returned units?
8. How are returned units distributed by return reason and product type?
9. How do return reasons vary across regions?

9.2 Average Return Rate

The average return rate was calculated to measure the percentage of sold products that were returned by customers.

Python Calculation

```
df_clean["Return Rate (%)"] = (
    df_clean["Units Returned"] /
    df_clean["Number of products sold"]
) * 100

avg_return_rate = (
    df_clean["Return Rate (%)"]
    .mean()
)

print(f"Average Return Rate: {avg_return_rate:.2f}%")
```

0.0s

```
Average Return Rate: 7.38%
```

SQL Validation

```
342  -- AVG Retuen Rate
343  SELECT
344      (SUM(Units_Returned) * 1.0 / SUM(Number_of_products_sold)) * 100 AS Average_Return_Rate
345  FROM Orders
346
```

92 % 15 0

Results Messages

	Average_Return_Rate
1	7.346101048800

Excel

Table name: Orders

Measure name: Return Rate

Description:

Formula: fx Check formula

```
=Sum(Orders[Units Returned])/Sum(Orders[Number of products sold])
```

SUM(ColumnName)

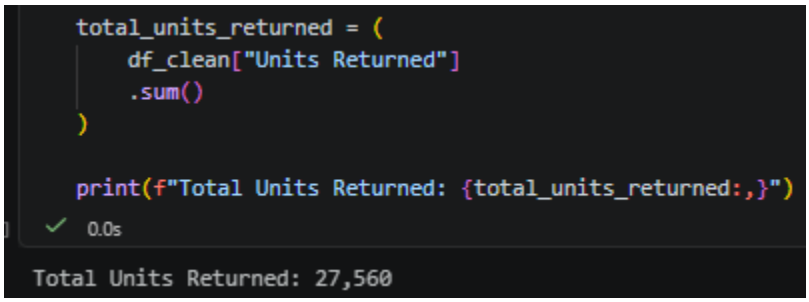
Return Rate
7.35%

The analysis shows that the average return rate is **7.38%**, indicating that a relatively small proportion of sold products were returned by customers.

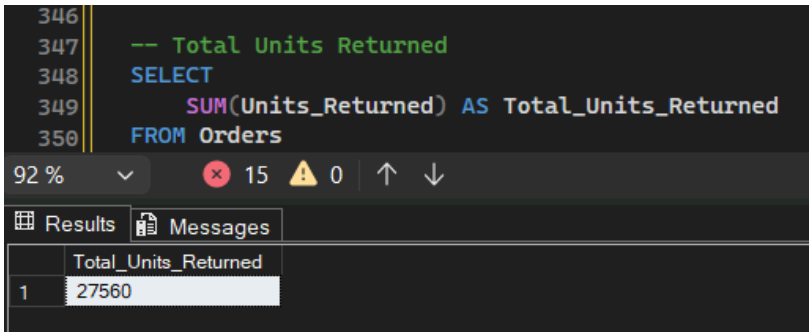
9.3 Total Units Returned

The total number of returned units was calculated to measure the overall volume of products returned by customers.

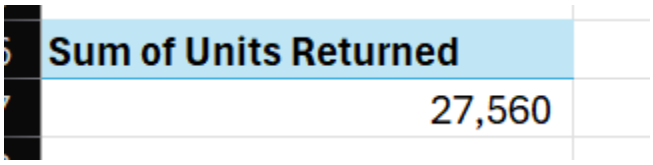
Python Calculation



SQL Validation



Excel



The analysis shows that a total of **27,560** units were returned, providing an overall measure of product .

9.4 Average Customer Satisfaction Score

The average customer satisfaction score was calculated to evaluate the overall satisfaction level of customers based on their ratings.

Python Calculation

```
avg_satisfaction_score = (
    df_clean["Customer Satisfaction Score"]
    .mean()
)

print(f"Average Customer Satisfaction Score: {avg_satisfaction_score:.2f}")
```

✓ 0.0s

```
Average Customer Satisfaction Score: 3.28
```

SQL Validation

```
352  -- AVG Customer Satisfaction Score
353  SELECT
354      AVG(Customer_Satisfaction_Score) AS Average_Satisfaction_Score
355  FROM Orders
356
```

92 % ⊗ 15 ⚠ 0 ↑ ↓

Results | Messages

	Average_Satisfaction_Score
1	3.28013333333333

Excel

9	Average of Customer Satisfaction Score
10	3.28

The analysis indicates that the average customer satisfaction score is **3.28**, reflecting a moderate level of customer satisfaction across all orders.

9.5 Total Estimated Return Loss

The total estimated return loss was calculated by multiplying the number of returned units by the product price for each order and then summing the estimated loss across all returned products.

Python Calculation

```
df_clean["Estimated Return Loss"] = (
    df_clean["Units Returned"] *
    df_clean["Price"]
)

total_return_loss = (
    df_clean["Estimated Return Loss"]
    .sum()
)

print(f"Total Estimated Return Loss: ${total_return_loss:,.2f}")
```

0.0s

```
Total Estimated Return Loss: $14,169,641.75
```

SQL Validation

```
356  -- Total Estimated Return Loss
357
358  SELECT
359      SUM(o.Units_Returned * p.Price) AS Total_Estimated_Return_Loss
360  FROM orders o
361  JOIN products p ON o.SKU = p.SKU
362
```

92 % 15 0

Results Messages

	Total_Estimated_Return_Loss
1	13435067.12

Excel

Table name: Orders

Measure name: Total Estimated Return Lose

Description:

Formula: fx Check formula

```
=SUM(Orders[Units Returned])*
  AVERAGE(Orders[Price])
```

2	Total Estimated Return Lose
3	14,183,960.9

The analysis estimates the total return loss at **$14,169,641.75**, representing the overall financial impact of returned products based on their selling prices.

9.6 Average Customer Satisfaction Score by Customer Demographics

The average customer satisfaction score was analyzed across different customer demographic groups to compare their overall satisfaction levels.

Python Calculation

```
gender_satisfaction = (
    df_clean
    .groupby("Customer demographics")["Customer Satisfaction Score"]
    .mean()
    .sort_values(ascending=False)
    .reset_index()
)

gender_satisfaction
```

✓ 0.0s

	Customer demographics	Customer Satisfaction Score
0	male	3.308083
1	non-binary	3.292484
2	female	3.237628

SQL Validation

```
447  SELECT
448      o.Customer_demographics,
449      AVG(o.Customer_Satisfaction_Score) AS avg_csat
450  FROM Orders o
451  GROUP BY
452      o.Customer_demographics
453  ORDER BY avg_csat ASC
454
455
```

92 % ✕ 15 ⚠ 0

Results | Messages

	Customer_demographics	avg_csat
1	female	3.23762781186094
2	non-binary	3.29248434237996
3	male	3.30808270676692

Excel

Row Labels	Average of Customer Satisfaction Score
Female	3.24
Male	3.31
Non-binary	3.29
Grand Total	**3.28**

The analysis shows that **Male** customers reported the highest average satisfaction score (**3.31**), followed by **Non-binary** customers (**3.29**), while **Female** customers recorded the lowest average score (**3.24**). Overall, the satisfaction levels are relatively similar across all demographic groups.

9.7 Return Rate & Estimated Return Loss by Region

The return rate and estimated return loss were analyzed across regions to identify the geographical areas with the highest return activity and financial impact.

Python Calculation

```python
region_returns = (
    df_clean
    .groupby("Region")
    .agg({
        "Units Returned": "sum",
        "Number of products sold": "sum",
        "Estimated Return Loss": "sum"
    })
)

region_returns["Return Rate (%)"] = (
    region_returns["Units Returned"] /
    region_returns["Number of products sold"]
) * 100

region_returns = (
    region_returns[
        ["Return Rate (%)", "Estimated Return Loss"]
    ]
    .sort_values(
        by="Estimated Return Loss",
        ascending=False
    )
)

region_returns
```

SQL Validation

```sql
SELECT
    o.Region,
    (SUM(o.Units_Returned) * 1.0 / NULLIF(SUM(o.Number_of_products_sold), 0)) * 100 AS Average_Return_Rate,
    SUM(o.Units_Returned * p.Price) AS Total_Estimated_Return_Loss
FROM Orders o
JOIN Products p ON o.SKU = p.SKU
GROUP BY o.Region
ORDER BY Average_Return_Rate DESC
```

92 % 15 0

Results Messages

	Region	Average_Return_Rate	Total_Estimated_Return_Loss
1	east	7.648436423700	2697280.55
2	west	7.447421289900	2741912.05
3	central	7.424763094800	2602535.59
4	south	7.366804085800	2664290.88
5	north	6.877024717400	2729048.05

Excel

Row Labels	Return Rate	Total Estimated Return Lose
Central	7.4%	2768659.64
East	7.6%	2934231.25
North	6.9%	2744409.48
South	7.4%	2873162.19
West	7.4%	2856400.96
Grand Total	**7.3%**	**14183960.88**

The analysis shows that the **North** region has the highest estimated return loss (**$2.91M**), while the **East** region records the highest return rate (**7.65%**). This indicates that although the North experiences the greatest financial loss, the East has the highest proportion of returned products.

9.8 Return Rate & Units Returned by Sales Channel

The total returned units and return rate were analyzed across different sales channels to evaluate their return performance.

Python Calculation

```python
sales_channel_returns = (
    df_clean
    .groupby("Sales Channel")
    .agg({
        "Units Returned": "sum",
        "Number of products sold": "sum"
    })
)

sales_channel_returns["Return Rate (%)"] = (
    sales_channel_returns["Units Returned"] /
    sales_channel_returns["Number of products sold"]
) * 100

sales_channel_returns = (
    sales_channel_returns[
        ["Units Returned", "Return Rate (%)"]
    ]
    .sort_values(
        by="Units Returned",
        ascending=False
    )
    .reset_index()
)
```

SQL Validation

```sql
--3.    Return Rate & Units Returned by Sales Channel ?
SELECT
    Sales_Channel,
    SUM(Units_Returned) AS Total_Units_Returned,
    (SUM(Units_Returned) * 1.0 / NULLIF(SUM(Number_of_products_sold), 0)) * 100 AS Return_Rate
FROM Orders
GROUP BY Sales_Channel
ORDER BY Return_Rate DESC
```

92 % 15 0

Results Messages

	Sales_Channel	Total_Units_Returned	Return_Rate
1	online	5788	7.691694352100
2	b2b	6069	7.537351432500
3	direct sales	5625	7.399563261300
4	wholesale	4983	7.097481768400
5	retail store	5095	6.963236299000

Excel

Row Labels	Sum of Units Returned	Return Rate
B2B	6,069	7.54%
Direct Sales	5,625	7.40%
Online	5,788	7.69%
Retail Store	5,095	6.96%
Wholesale	4,983	7.10%
Grand Total	**27,560**	**7.35%**

The analysis shows that the **B2B** sales channel has the highest number of returned units (**6,069**), while the **Online** channel records the highest return rate (**7.69%**). This indicates that although B2B generates the largest return volume, Online has the highest proportion of returned products.

9.9 Units Returned by Return Reason and Product Type

Returned units were analyzed by both return reason and product type to identify which products are most frequently returned for each reason.

Python Calculation

```python
return_reason_product = (
    df_clean
    .pivot_table(
        values="Units Returned",
        index="Return Reason",
        columns="Product type",
        aggfunc="sum",
        fill_value=0
    )
)

return_reason_product = return_reason_product.drop(index="no return")

return_reason_product
```

SQL Validation

```sql
--4.    Units Returned by Return Reason and Product Type
SELECT
    o.Return_Reason,
    p.Product_Type,
    SUM(o.Units_Returned) AS Total_Units_Returned
FROM Orders o
JOIN Products p ON o.SKU = p.SKU
WHERE o.Return_Reason <> 'no return'
GROUP BY o.Return_Reason, p.Product_Type
ORDER BY o.Return_Reason, Total_Units_Returned DESC
```

Excel

Sum of Units Returned	Column Labels					
Row Labels	Headphones	Laptops	Smartphones	Smartwatches	Tablets	Grand Total
Changed Mind	867	1,028	1,482	600	1,143	5,120
Damaged in Shipping	781	859	901	518	732	3,791
Defective	1,974	1,939	2,115	1,417	2,143	9,588
Not as Described	937	917	1,336	1,027	833	5,050
Wrong Item	610	679	1,323	661	738	4,011
Grand Total	**5,169**	**5,422**	**7,157**	**4,223**	**5,589**	**27,560**

The analysis shows that **Defective** is the most common return reason across all product types. **Smartphones** and **Tablets** recorded the highest number of returned units in most return categories, while **Smartwatches** generally had the lowest number of returns.

9.10 Return Reasons Across Regions

Return reasons were analyzed across different regions to identify the most common causes of product returns in each geographical area.

Python Calculation

```python
return_reason_region = (
    df_clean[df_clean["Return Reason"] != "no return"]
    .pivot_table(
        values="Units Returned",
        index="Region",
        columns="Return Reason",
        aggfunc="sum",
        fill_value=0
    )
)

return_reason_region
```

SQL Validation

```sql
--2. Return Reasons Across Regions
 SELECT
    Region,
    Return_Reason,
    SUM(Units_Returned) AS Total_Units_Returned
FROM Orders
WHERE Return_Reason <> 'no return'
GROUP BY Region, Return_Reason
ORDER BY Region, Total_Units_Returned DESC
```

Excel

Sum of Units Returned	Column Labels					
Row Labels	Changed Mind	Damaged in Shipping	Defective	Not as Described	Wrong Item	Grand Total
Central	957	855	2,116	910	866	5,704
East	995	737	2,109	979	732	5,552
North	1,027	637	1,929	1,200	599	5,392
South	1,210	490	1,791	1,128	992	5,611
West	931	1,072	1,643	833	822	5,301
Grand Total	**5,120**	**3,791**	**9,588**	**5,050**	**4,011**	**27,560**

The analysis shows that **Defective** is the leading return reason across all regions. The **Central** region recorded the highest number of defective returns (**2,116**), while the **West** region had the highest number of returns due to **Damaged in Shipping** (**1,072**). Return patterns are generally consistent across regions, with

9.11 Domain Summary

The Customer Experience & Returns analysis evaluated customer satisfaction, product returns, return losses, and return patterns across multiple business dimensions. The analysis measured key performance indicators such as average return rate, total returned units, customer satisfaction score, and estimated return loss. It also compared return behavior across customer demographics, regions, sales channels, product types, and return reasons.

Overall, the analysis identified the primary drivers of product returns and highlighted areas with higher financial impact, providing valuable insights for improving customer satisfaction and reducing return-related losses.

Key Insights

- The average return rate across all orders is **7.38%**.
- A total of **27,560** units were returned.
- The average customer satisfaction score is **3.28**.
- The estimated financial impact of returned products is approximately **$14.17M**.
- Male customers reported the highest average satisfaction score (**3.31**).
- The **North** region recorded the highest estimated return loss, while the **East** region showed the highest return rate.
- The **B2B** sales channel generated the highest number of returned units, whereas the **Online** channel recorded the highest return rate.
- **Defective products** were the most common reason for returns across all product categories and regions.

Business Value

This analysis helps identify return trends, evaluate customer satisfaction, and estimate the financial impact of product returns. These insights support decision-making for improving product quality, enhancing customer experience, optimizing sales channels, and reducing return-related costs.

---

10. Domain 3: Suppliers Performance Dashboard

10.1 Overview

This domain evaluates supplier performance by analyzing product quality, supplier reliability, manufacturing costs, profitability, return rates, and defect rates. The analysis aims to identify the best-performing suppliers, evaluate supplier efficiency, and measure the impact of supplier quality on business performance.

Business Questions

1. What is the average supplier quality score?
2. What is the average supplier reliability score?
3. What is the average return rate?
4. What is the average defect rate?
5. How does manufacturing cost compare with quality?
6. Which supplier generates the highest profit and quality score?
7. Which supplier has the highest value score?
8. Which supplier has the highest defect rate?
9. How does the return rate vary across suppliers?

10.2 Average Supplier Quality Score

The average supplier quality score was calculated to evaluate the overall quality performance of all suppliers.

Python Calculation

```
avg_quality_score = (
    df_clean["Quality Score"]
    .mean()
)

print(f"Average Supplier Quality Score: {avg_quality_score:.2f}")
```

✓ 0.0s

```
Average Supplier Quality Score: 50.23
```

SQL Validation

```
-- domain 3 (supplier performance)
-- Average Supplier Quality Score
SELECT
    AVG(Quality_Score) AS Average_Supplier_Quality_Score
FROM
    Products
```

92 % 15 0

Results Messages

	Average_Supplier_Quality_Score
1	50.2321333333333

Excel

	Average of Quality Score
2	
3	50.23

The analysis shows that the average supplier quality score is **50.23**, indicating a moderate overall quality level across all suppliers.

10.3 Average Supplier Reliability Score

The average supplier reliability score was calculated to evaluate the overall reliability and consistency of suppliers in fulfilling orders.

Python Calculation

```
avg_reliability_score = (
    df_clean["Supplier Reliability Score"]
    .mean()
) * 100

print(f"Average Supplier Reliability Score: {avg_reliability_score:.2f}%")
```

0.0s

```
Average Supplier Reliability Score: 65.30%
```

The analysis shows that the average supplier reliability score is **65.30%**, indicating a good overall level of supplier reliability.

SQL Validation

```sql
-- Average Supplier Reliability Score
SELECT
    AVG(Avg_Reliability_Score) AS Average_Supplier_Reliability_Score
FROM
    Suppliers
```

92 % 15 0

Results Messages

	Average_Supplier_Reliability_Score
1	0.6530678

Excel

Average of Avg. Reliability Score
65.32%

10.4 Average Return Rate

The average return rate was calculated to measure the percentage of sold products that were returned across all suppliers.

Python Calculation

```python
avg_return_rate = (
    df_clean["Return Rate (%)"]
    .mean()
)

print(f"Average Return Rate: {avg_return_rate:.2f}%")
```

SQL Validation

```sql
-- AVG Retuen Rate
SELECT
    (SUM(Units_Returned) * 1.0 / SUM(Number_of_products_sold)) * 100 AS Average_Return_Rate
FROM Orders
```

92 % 15 0

Results Messages

	Average_Return_Rate
1	7.346101048800

Excel

Return Rate
7.35%

The analysis shows that the average return rate is **7.38%**, indicating that approximately 7 out of every 100 sold products were returned.

10.5 Average Defect Rate

The average defect rate was calculated to evaluate the percentage of defective products supplied across all suppliers.

Python Calculation

```python
avg_defect_rate = (
    df_clean["Defect rates"]
    .mean()
)

print(f"Average Defect Rate: {avg_defect_rate:.2f}%")
```

SQL Validation

```sql
-- Average Defect Rate
SELECT
    AVG(Defect_rates) AS Average_Defect_Rate
FROM
    Products
```

92 % 15 0

Results Messages

	Average_Defect_Rate
1	4.97664

Excel

Average of Defect rates1
4.98

The analysis shows that the average defect rate is **4.98%**, indicating that approximately 5% of supplied products were identified as defective.

10.6 Manufacturing Cost vs Quality Score

The average manufacturing cost and average quality score were compared across suppliers to evaluate the relationship between production cost and product quality.

Python Calculation

```python
manufacturing_quality = (
    df_clean
    .groupby("Supplier name")
    .agg({
        "Manufacturing costs": "mean",
        "Quality Score": "mean"
    })
    .round(2)
    .sort_values(
        by="Quality Score",
        ascending=False
    )
    .reset_index()
)
```

SQL Validation

```sql
--4.    Which suppliers provide high quality at optimal cost?
SELECT
    s.Supplier_name,
    AVG(p.Quality_Score) AS avg_quality,
    AVG(p.Manufacturing_costs) AS avg_cost
FROM Orders o
JOIN Suppliers s
    ON o.SupplierID = s.SupplierID
JOIN Products p
    ON o.SKU = p.SKU
GROUP BY s.Supplier_name
ORDER BY avg_quality DESC, avg_cost ASC
```

92 % 15 0

Results Messages

	Supplier_name	avg_quality	avg_cost
1	supplier 4	51.6436184210527	218.617434
2	supplier 1	51.2851748251748	213.550979
3	supplier 3	50.7699668874172	231.397483
4	supplier 2	49.4693624161074	235.549563
5	supplier 5	48.0857419354839	210.140451

Excel

Row Labels	Average of Manufacturing costs	Average of Quality Score1
Supplier 1	213.6	51.3
Supplier 2	235.5	49.5
Supplier 3	231.4	50.8
Supplier 4	218.6	51.6
Supplier 5	210.1	48.1
Grand Total	**221.8**	**50.2**

Chart Area

The analysis shows that **Supplier 4** achieved the highest average quality score (**51.64**) with an average manufacturing cost of **218.62**. In contrast, **Supplier 2** recorded the highest manufacturing cost (**235.55**) while delivering a lower quality score (**49.47**), indicating that higher production costs do not necessarily result in better product quality.

10.7 Supplier Profit & Quality Score

Description

Supplier performance was evaluated by comparing total profit with the average quality score for each supplier.

Python Calculation

```
supplier_profit_quality = (
    df_clean
    .groupby("Supplier name")
    .agg({
        "Profit": "sum",
        "Quality Score": "mean"
    })
    .round(2)
    .sort_values(
        by="Profit",
        ascending=False
    )
    .reset_index()
)
```

SQL Validation

```sql
--2.    Supplier Profit & Quality Score
SELECT
    s.Supplier_name,
    SUM(o.Profit) AS Total_Profit,
    AVG(p.Quality_Score) AS Average_Quality_Score
FROM
    Suppliers s
JOIN
    Orders o ON s.SupplierID = o.SupplierID
JOIN
    Products p ON o.SKU = p.SKU
GROUP BY
    s.Supplier_name
ORDER BY
    Total_Profit DESC
```

92 % 15 0

Results Messages

	Supplier_name	Total_Profit	Average_Quality_Score
1	supplier 2	18532887.23	49.4693624161074
2	supplier 5	17628774.88	48.0857419354839
3	supplier 3	16744544.22	50.7699668874172
4	supplier 1	16676678.51	51.2851748251748
5	supplier 4	16043915.76	51.6436184210527

Excel

Row Labels	Sum of Profit	Average of Quality Score1
Supplier 1	$ 16,676,678.49	51.3
Supplier 2	$ 18,532,887.30	49.5
Supplier 3	$ 16,744,544.24	50.8
Supplier 4	$ 16,043,915.75	51.6
Supplier 5	$ 17,628,774.89	48.1
Grand Total	**$ 85,626,800.67**	**50.2**

The analysis shows that **Supplier 2** generated the highest total profit (**$18.53M**), while **Supplier 4** achieved the highest average quality score (**51.64**). This indicates that the most profitable supplier is not necessarily the supplier with the highest product quality.

10.8 Supplier Value Score

A supplier value score was calculated by combining the average quality score and supplier reliability score, then dividing the result by the average defect rate to evaluate the overall supplier performance.

Python Calculation

```python
supplier_value_score = (
    df_clean.groupby("Supplier name")
    .agg({
        "Quality Score":"mean",
        "Supplier Reliability Score":"mean",
        "Defect rates":"mean"
    })
)

supplier_value_score["Value Score"] = (
    supplier_value_score["Quality Score"] *
    supplier_value_score["Supplier Reliability Score"] /
    supplier_value_score["Defect rates"]
)

supplier_value_score = (
    supplier_value_score[["Value Score"]]
    .round(3)
    .sort_values(by="Value Score", ascending=False)
    .reset_index()
)
```

SQL Validation

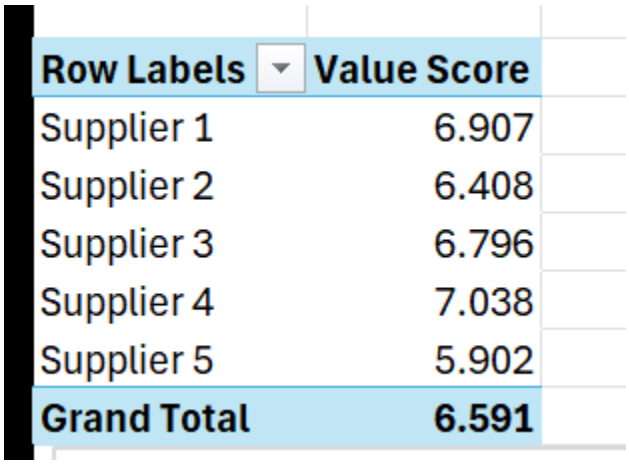
```sql
--3.    Supplier Value Score
SELECT
    Supplier_name,
    ROUND((Avg_Quality * Avg_Reliability) / Avg_Defect, 3) AS Value_Score
FROM (
    SELECT
        s.Supplier_name,
        AVG(p.Quality_Score) AS Avg_Quality,
        AVG(s.Avg_Reliability_Score) AS Avg_Reliability,
        AVG(p.Defect_rates) AS Avg_Defect
    FROM Suppliers s
    JOIN Orders o ON s.SupplierID = o.SupplierID
    JOIN Products p ON o.SKU = p.SKU
    GROUP BY s.Supplier_name
) AS AggregatedData
ORDER BY Value_Score DESC
```

92 % ✕ 15 ⚠ 0

Results | Messages

	Supplier_name	Value_Score
1	supplier 4	7.039
2	supplier 1	6.903
3	supplier 3	6.795
4	supplier 2	6.407
5	supplier 5	5.903

Excel



Row Labels	Value Score
Supplier 1	6.907
Supplier 2	6.408
Supplier 3	6.796
Supplier 4	7.038
Supplier 5	5.902
Grand Total	**6.591**

The analysis shows that **Supplier 4** achieved the highest value score (**7.038**), indicating the best overall balance between quality, reliability, and defect rate. **Supplier 5** recorded the lowest value score (**5.902**), suggesting room for performance improvement.

10.9 Average Defect Rate by Supplier

The average defect rate was calculated for each supplier to identify suppliers with the highest proportion of defective products.

Python Calculation

```
supplier_defect_rate = (
    df_clean.groupby("Supplier name")["Defect rates"]
    .mean()
    .sort_values(ascending=False)
    .reset_index()
)
```

SQL Validation

```
480
481  --6.    Avg defect rates affect each supplier
482  SELECT
483      s.Supplier_name,
484      AVG(p.Defect_rates) AS average_defect_rate
485  FROM
486      Suppliers s
487  JOIN
488      Orders o ON s.SupplierID = o.SupplierID
489  JOIN
490      Products p ON o.SKU = p.SKU
491  GROUP BY
492      s.Supplier_name
493  ORDER BY
494      average_defect_rate DESC
495
```

92 % 15 0

Results Messages

	Supplier_name	average_defect_rate
1	supplier 5	5.19122580645162
2	supplier 2	5.05281879194631
3	supplier 3	4.92291390728476
4	supplier 1	4.87136363636364
5	supplier 4	4.83555921052632

Excel

Row Labels	Average of Defect rates1
Supplier 1	4.87
Supplier 2	5.05
Supplier 3	4.92
Supplier 4	4.84
Supplier 5	5.19
Grand Total	**4.98**

The analysis shows that **Supplier 5** recorded the highest average defect rate (**5.19%**), followed by **Supplier 2** (**5.05%**). In contrast, **Supplier 4** achieved the lowest defect rate (**4.84%**), indicating better product quality and manufacturing consistency.

10.10 Average Return Rate by Supplier

The average return rate was calculated for each supplier to compare return performance across suppliers.

Python Calculation

```
supplier_return_rate = (
    df_clean.groupby("Supplier name")["Return Rate (%)"]
    .mean()
    .round(2)
    .sort_values(ascending=False)
    .reset_index()
)
```

SQL Validation

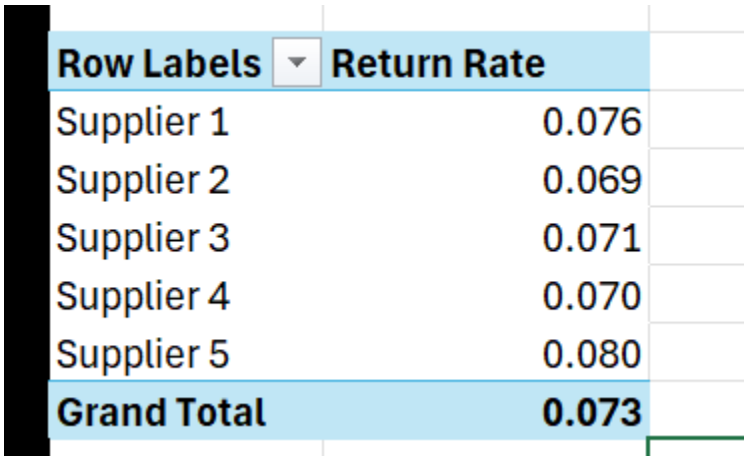
```
--5.    Is there a correlation between suppliers and high return rates?
SELECT
    s.Supplier_name,
    SUM(o.Units_Returned) * 1.0 / SUM(o.Number_of_products_sold) AS return_rate
FROM Orders o
JOIN Suppliers s
    ON o.SupplierID = s.SupplierID
GROUP BY s.Supplier_name
ORDER BY return_rate DESC
```

92 % 15 0

Results Messages

	Supplier_name	return_rate
1	supplier 5	0.079986595865
2	supplier 1	0.076064096823
3	supplier 3	0.071368997974
4	supplier 4	0.070427718561
5	supplier 2	0.069224920971

Excel



Row Labels	Return Rate
Supplier 1	0.076
Supplier 2	0.069
Supplier 3	0.071
Supplier 4	0.070
Supplier 5	0.080
Grand Total	**0.073**

The analysis shows that **Supplier 5** recorded the highest average return rate (**7.72%**), followed by **Supplier 1** (**7.45%**). Meanwhile, **Supplier 2** achieved the lowest return rate (**7.18%**), indicating better return performance compared to the other suppliers.

10.11 Domain Summary

The Suppliers Performance analysis evaluated supplier efficiency by examining quality, reliability, defect rates, return rates, manufacturing costs, profitability, and overall supplier value. The analysis compared suppliers across multiple performance indicators to identify top-performing suppliers and areas requiring improvement.

Overall, the analysis provides a comprehensive assessment of supplier performance and supports supplier evaluation based on quality, operational efficiency, and financial contribution.

Key Insights

- The average supplier quality score is **50.23**.
- The average supplier reliability score reached **65.30%**.
- The overall average return rate is **7.38%**.
- The average defect rate across suppliers is **4.98%**.
- **Supplier 4** achieved the highest quality score while maintaining a moderate manufacturing cost.
- **Supplier 2** generated the highest total profit among all suppliers.
- **Supplier 4** recorded the highest supplier value score (**7.038**), reflecting the best balance between quality, reliability, and defect rate.
- **Supplier 5** had the highest defect rate (**5.19%**) and the highest average return rate (**7.72%**).

Business Value

This analysis provides a comprehensive evaluation of supplier performance by combining operational, quality, and financial metrics. The findings support supplier selection, performance monitoring, quality improvement initiatives, and strategic sourcing decisions, helping the organization strengthen supplier relationships while reducing quality issues and return-related costs.

---

11. Domain 4: Inventory

11.1 Overview

This domain analyzes inventory performance by evaluating production volume, stock availability, inventory gaps, stockout risks, product profitability, and SKU performance. The analysis aims to optimize inventory management, identify stock shortages, and evaluate product performance across different inventory metrics.

Business Questions

1. What is the total production volume?
2. What is the total number of units sold?
3. What is the total stock available?
4. What is the inventory gap?
5. How many products are available?
6. How are SKUs distributed by stockout risk?
7. Which product type has the highest production volume?
8. Which product type has the largest inventory gap?
9. Which products have the highest profit margin?
10. Which SKUs are the top-selling products?
11. How does profit margin vary across price ranges?
12. Which product type has the highest defect rate?

11.2 Total Production Volume

The total production volume was calculated to measure the overall number of units produced across all products.

```
total_production = df_clean["Production volumes"].sum()

print(f"Total Production Volume: {total_production:,.0f}")
```

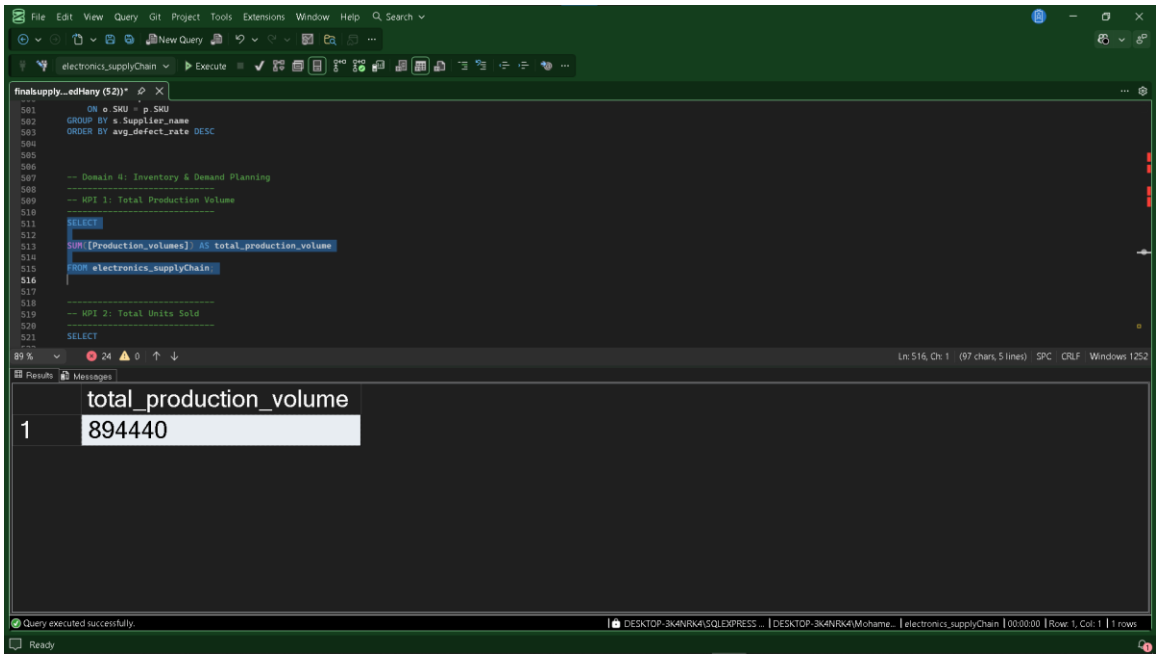
✓ 0.0s

```
Total Production Volume: 894,440
```

Excel

Sum of Production volumes
894,440

SQL Validation



The analysis shows that the total production volume is **894,440** units, representing the overall manufacturing output available for inventory management.

11.3 Total Units Sold

The total number of units sold was calculated to measure the overall sales volume across all products.

```
total_units_sold = df_clean["Number of products sold"].sum()

print(f"Total Units Sold: {total_units_sold:,.0f}")
```

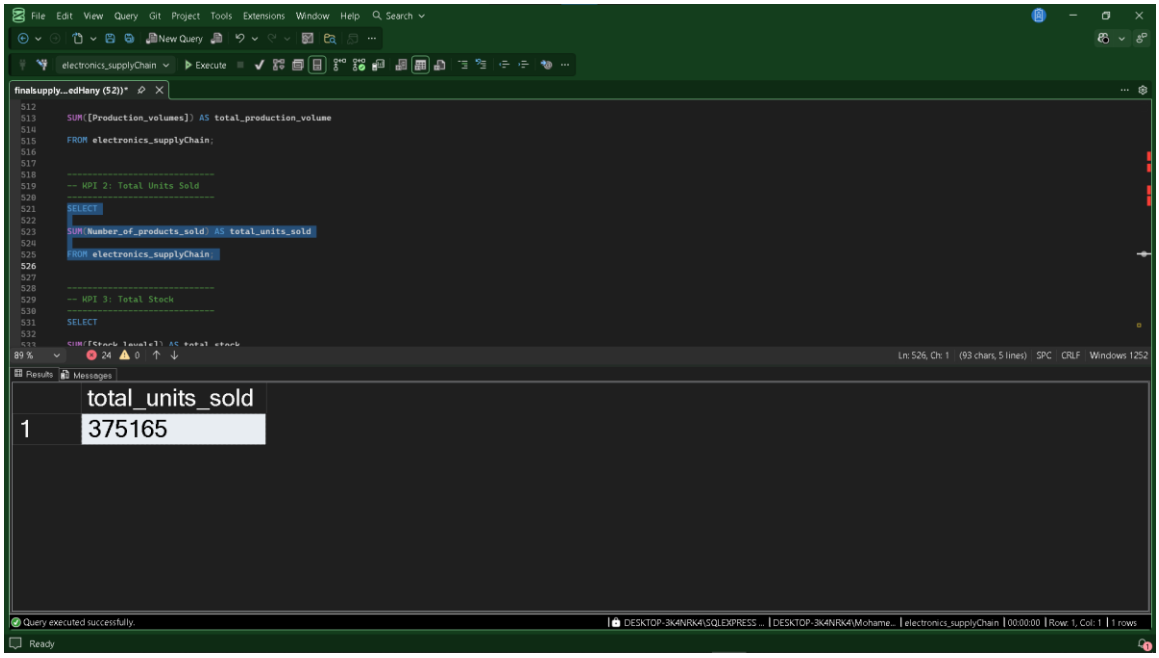
✓ 0.0s

```
Total Units Sold: 375,165
```

Excel

Sum of Number of products sold
375,165

SQL Validation



The analysis shows that a total of **375,165** units were sold, representing the overall sales volume during the analyzed period.

11.4 Total Stock Available

The total stock available was calculated to determine the overall inventory currently available across all products.

```
total_stock = df_clean["Stock levels"].sum()

print(f"Total Stock Available: {total_stock:,.0f}")
```

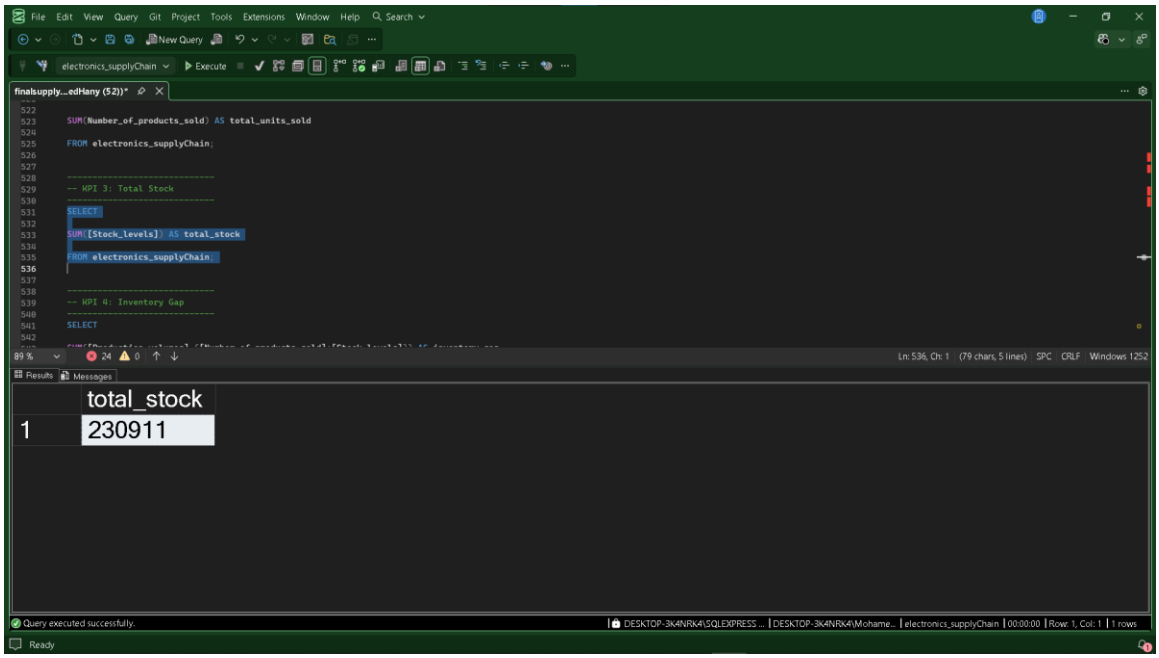
✓ 0.0s

```
Total Stock Available: 230,911
```

Excel

Sum of Stock levels
230,911

SQL Validation



The analysis shows that the total stock available is **230,911** units, representing the inventory currently available to support future customer demand.

11.5 Inventory Gap

The inventory gap was calculated by subtracting the total units sold and available stock from the total production volume to measure the remaining inventory difference.

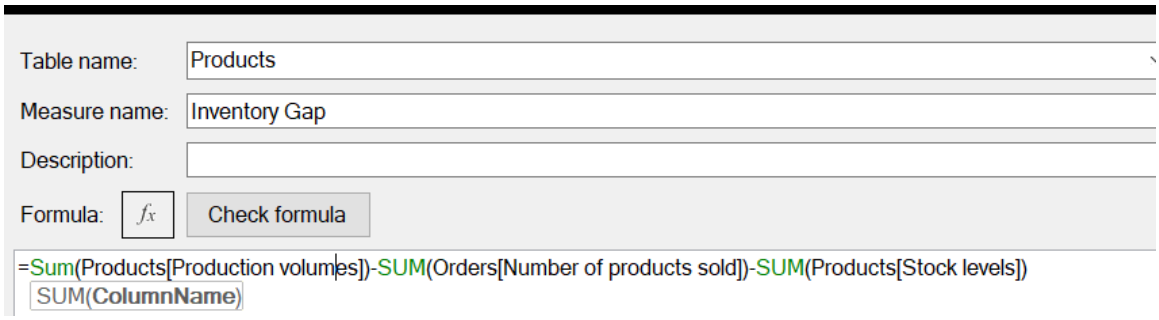
```
inventory_gap = (
    df_clean["Production volumes"].sum()
    - (
        df_clean["Number of products sold"].sum()
        + df_clean["Stock levels"].sum()
    )
)

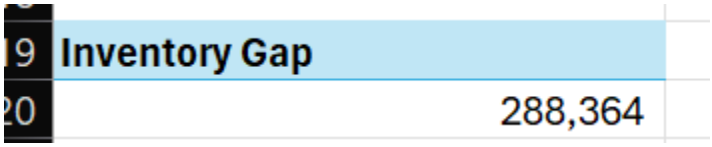
print(f"Inventory Gap: {inventory_gap:,.0f}")
```

✓ 0.0s

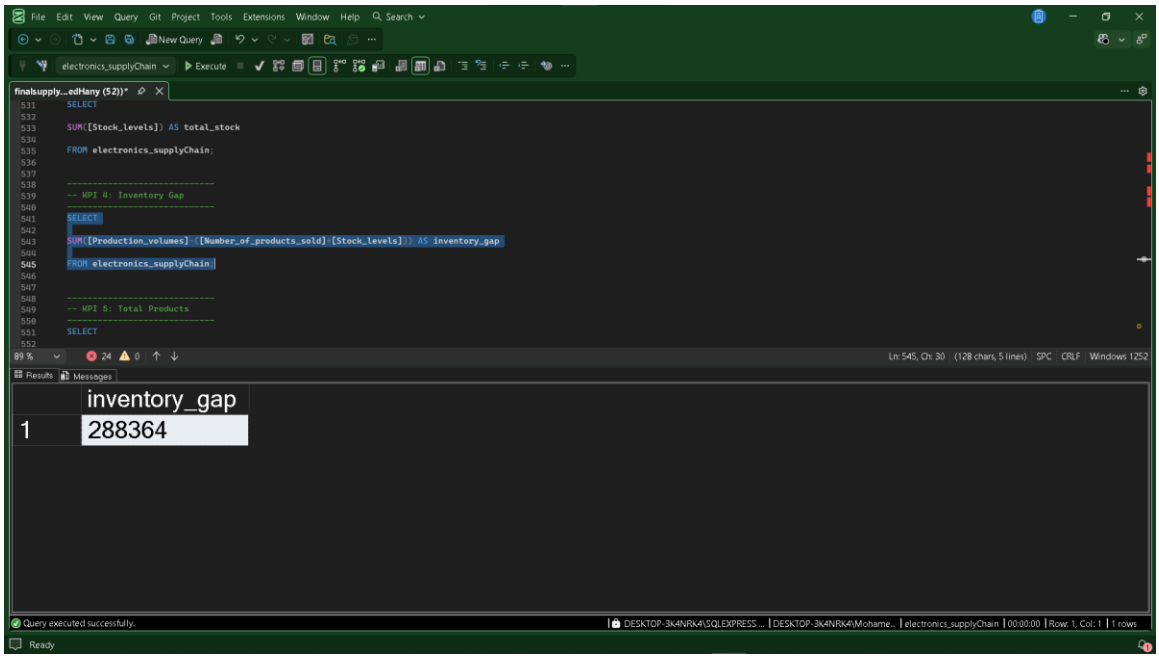
```
Inventory Gap: 288,364
```

Excel





SQL Validation



The analysis shows that the inventory gap is **288,364** units, indicating the difference between total production and the combined quantity of sold and available stock.

11.6 Number of Products

The total number of unique products was calculated by counting the distinct SKUs available in the dataset.

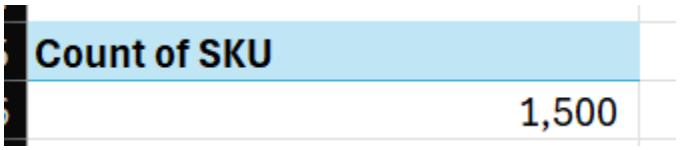
```
total_products = df_clean["SKU"].nunique()

print(f"Total Products: {total_products:,}")
```

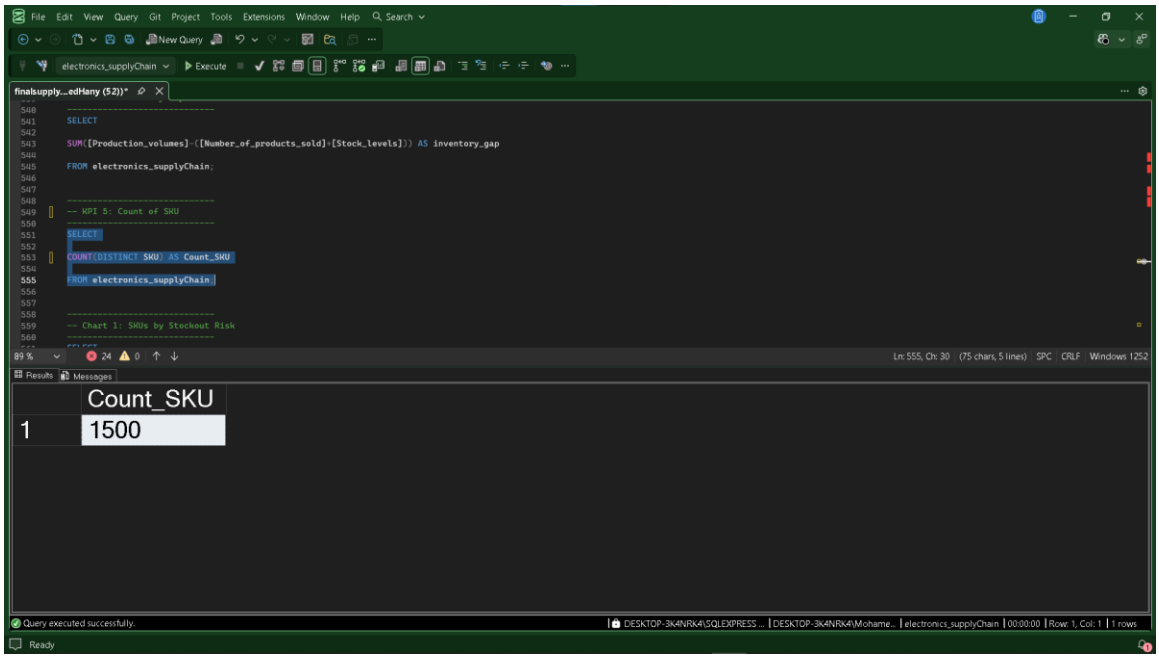
✓ 0.0s

```
Total Products: 1,500
```

Excel



SQL Validation



The analysis shows that the inventory consists of **1,500** unique products, representing the complete product portfolio available for analysis.

11.7 SKU Distribution by Stockout Risk

Products were classified based on their stockout status to determine the percentage of SKUs that are at risk of stock shortages.

```
df_clean["Stockout Flag"] = (
    df_clean["Stockout Risk"]
    .map({"Yes": "Risk", "No": "No Risk"})
)

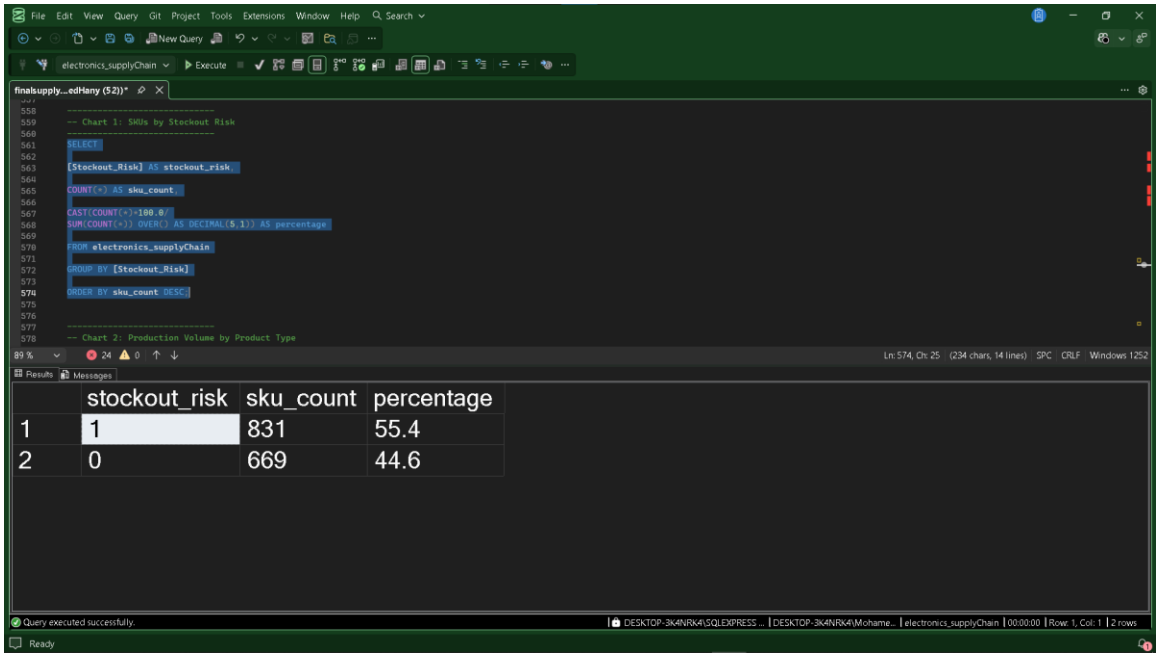
stockout_distribution = (
    df_clean.groupby("Stockout Flag")["SKU"]
    .count()
    .reset_index(name="SKUs")
)

stockout_distribution["Percentage (%)"] = (
    stockout_distribution["SKUs"] /
    stockout_distribution["SKUs"].sum() * 100
).round(1)
```

Excel

Row Labels	Count of SKU
No Risk	44.60%
Risk	55.40%
Grand Total	**100.00%**

SQL Validation



The analysis shows that **55.4%** of SKUs are classified as **Risk**, while **44.6%** are classified as **No Risk**, indicating that more than half of the products have a potential stockout risk.

11.8 Production Volume by Product Type

The percentage of total production volume was calculated for each product type to identify the products contributing most to overall production.

```
production_by_product = (
    df_clean.groupby("Product type")["Production volumes"]
    .sum()
    .reset_index()
)

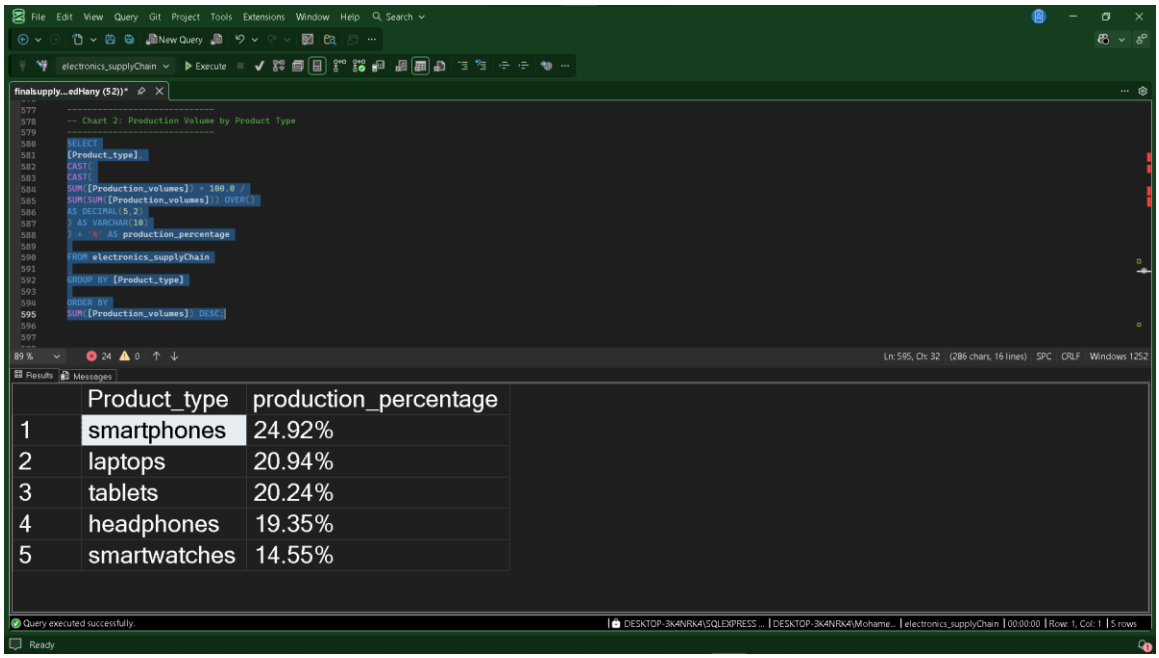
production_by_product["Percentage (%)"] = (
    production_by_product["Production volumes"] /
    production_by_product["Production volumes"].sum() * 100
).round(2)

production_by_product = (
    production_by_product[["Product type", "Percentage (%)"]]
    .sort_values(by="Percentage (%)", ascending=False)
    .reset_index(drop=True)
)
```

Excel

Row Labels	Sum of Production volumes
Headphones	19.35%
Laptops	20.94%
Smartphones	24.92%
Smartwatches	14.55%
Tablets	20.24%
Grand Total	**100.00%**

SQL Validation



```
-- Chart 2: Production Volume by Product Type
SELECT
[Product_type],
CAST(
CAST(
SUM([Production_volumes]) * 100.0 /
SUM(SUM([Production_volumes])) OVER()
AS DECIMAL(5,2))
) AS VARCHAR(10)
) + '%' AS production_percentage

FROM electronics_supplyChain

GROUP BY [Product_type]

ORDER BY
SUM([Production_volumes]) DESC;
```

	Product_type	production_percentage
1	smartphones	24.92%
2	laptops	20.94%
3	tablets	20.24%
4	headphones	19.35%
5	smartwatches	14.55%

The analysis shows that **Smartphones** account for the largest share of production (**24.92%**), followed by **Laptops (20.94%)** and **Tablets (20.24%)**, while **Smartwatches** represent the smallest production share (**14.55%**).

11.9 Inventory Gap & Stock Level by Product Type

Inventory gap and available stock were compared across product types to evaluate inventory availability and identify products with the largest inventory shortages.

```
inventory_analysis = (
    df_clean.groupby("Product type")
    .agg({
        "Production volumes":"sum",
        "Number of products sold":"sum",
        "Stock levels":"sum"
    })
)

inventory_analysis["Inventory Gap"] = (
    inventory_analysis["Production volumes"] -
    inventory_analysis["Number of products sold"] -
    inventory_analysis["Stock levels"]
)

inventory_analysis = (
    inventory_analysis[["Inventory Gap","Stock levels"]]
    .sort_values(by="Inventory Gap", ascending=False)
    .reset_index()
)
```

Excel

Row Labels	Sum of Stock levels	Inventory Gap
Headphones	42,885	59,786
Laptops	50,292	62,450
Smartphones	55,776	72,829
Smartwatches	35,417	36,035
Tablets	46,541	57,264
Grand Total	**230,911**	**288,364**

The analysis shows that **Smartphones** have the largest inventory gap (**72,829** units) while also maintaining the highest stock level (**55,776** units). In contrast, **Smartwatches** recorded the smallest inventory gap (**36,035** units) and the lowest stock level (**35,417** units).

11.10 Profit Margin & Units Sold by Product Type

The total units sold and profit margin were compared across product types to evaluate both sales performance and profitability.

```
product_profit = (
    df_clean.groupby("Product type")
    .agg({
        "Number of products sold":"sum",
        "Revenue generated":"sum",
        "Profit":"sum"
    })
)

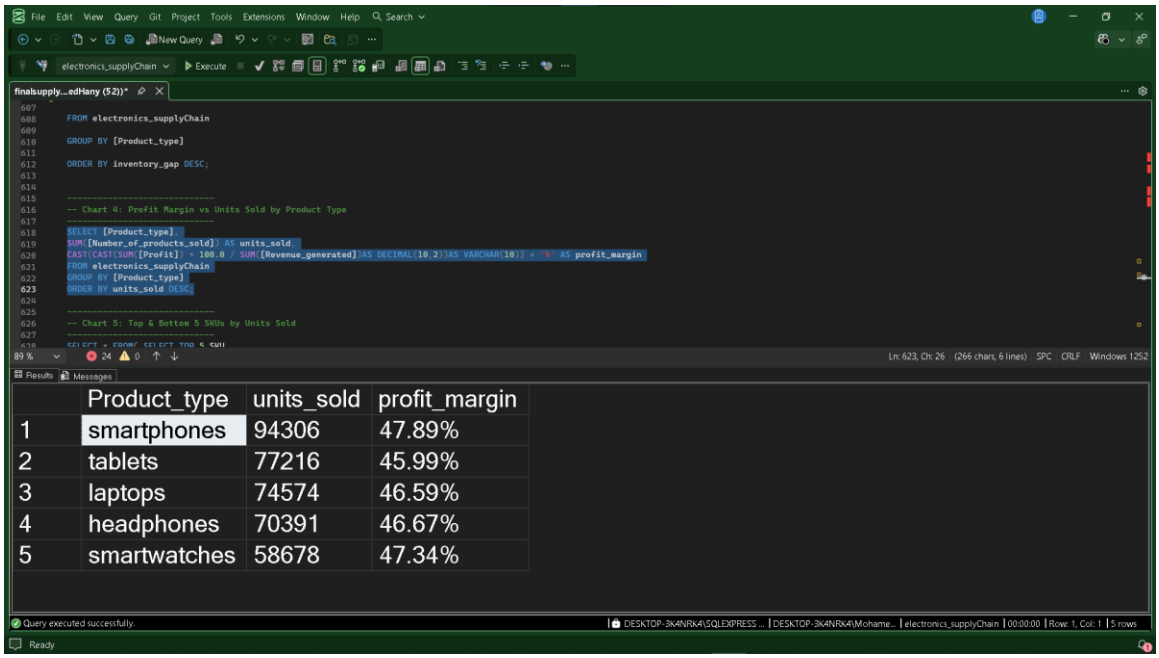
product_profit["Profit Margin (%)"] = (
    product_profit["Profit"] /
    product_profit["Revenue generated"] * 100
)

product_profit = (
    product_profit[["Number of products sold","Profit Margin (%)"]]
    .round(2)
    .sort_values(by="Number of products sold", ascending=False)
    .reset_index()
)
```

Excel

Row Labels	Profit Margin	Sum of Number of products sold
Headphones	46.67%	70,391
Laptops	46.59%	74,574
Smartphones	47.89%	94,306
Smartwatches	47.34%	58,678
Tablets	45.99%	77,216
Grand Total	**46.84%**	**375,165**

SQL Validation



```
FROM electronics_supplyChain

GROUP BY [Product_type]

ORDER BY inventory_gap DESC;


-----------------------------
-- Chart 4: Profit Margin vs Units Sold by Product Type
-----------------------------
SELECT [Product_type],
SUM([Number_of_products_sold]) AS units_sold,
CAST(CAST(SUM([Profit]) * 100.0 / SUM([Revenue_generated])AS DECIMAL(10,2))AS VARCHAR(10)) + '%' AS profit_margin
FROM electronics_supplyChain
GROUP BY [Product_type]
ORDER BY units_sold DESC;

-----------------------------
-- Chart 5: Top & Bottom 5 SKUs by Units Sold
-----------------------------
```

	Product_type	units_sold	profit_margin
1	smartphones	94306	47.89%
2	tablets	77216	45.99%
3	laptops	74574	46.59%
4	headphones	70391	46.67%
5	smartwatches	58678	47.34%

The analysis shows that **Smartphones** achieved the highest sales volume (**94,306** units) and the highest profit margin (**47.89%**). Although **Tablets** ranked second in units sold (**77,216**), they recorded the lowest profit margin (**45.99%**) among all product types.

11.11 Top & Bottom Selling SKUs

The top five and bottom five SKUs were identified based on the total number of units sold to highlight the best- and worst-performing products.

```python
top_skus = (
    df_clean[["SKU", "Number of products sold"]]
    .sort_values(by="Number of products sold", ascending=False)
    .head(5)
)

bottom_skus = (
    df_clean[["SKU", "Number of products sold"]]
    .sort_values(by="Number of products sold")
    .head(5)
)

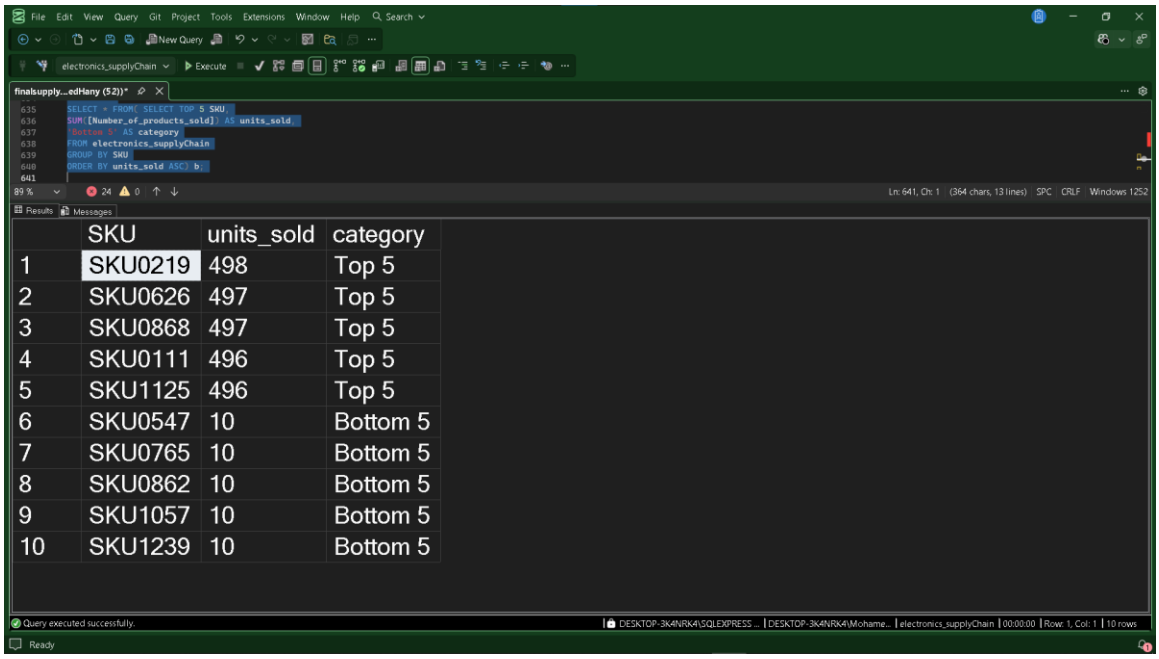
top_bottom_skus = (
    pd.concat([top_skus, bottom_skus])
    .reset_index(drop=True)
)
```

Excel

Row Labels	Sum of Number of products sold
SKU0219	498
SKU0868	497
SKU0626	497
SKU1125	496
SKU0111	496
Grand Total	**2484**

Row Labels	Sum of Number of products sold
SKU0547	10
SKU0765	10
SKU0862	10
SKU1057	10
SKU1239	10
Grand Total	**50**

SQL Validation



	SKU	units_sold	category
1	SKU0219	498	Top 5
2	SKU0626	497	Top 5
3	SKU0868	497	Top 5
4	SKU0111	496	Top 5
5	SKU1125	496	Top 5
6	SKU0547	10	Bottom 5
7	SKU0765	10	Bottom 5
8	SKU0862	10	Bottom 5
9	SKU1057	10	Bottom 5
10	SKU1239	10	Bottom 5

The analysis shows that **SKU0219** recorded the highest sales with **498** units, followed by **SKU0626** and **SKU0868** with **497** units each. On the other hand, **SKU1057**, **SKU0765**, **SKU0547**, **SKU1239**, and **SKU0862** were the lowest-selling products, each recording only **10** units sold.

11.12 Profit Margin by Price Range

Products were grouped into different price ranges to compare profitability across pricing segments.

```python
df_clean["Price Range"] = pd.cut(
    df_clean["Price"],
    bins=[0, 400, 800, 1200, float("inf")],
    labels=["0-400", "400-800", "800-1200", "1200+"]
)

price_margin = (
    df_clean.groupby("Price Range")
    .agg({
        "Revenue generated":"sum",
        "Profit":"sum"
    })
)

price_margin["Profit Margin (%)"] = (
    price_margin["Profit"] /
    price_margin["Revenue generated"] * 100
)

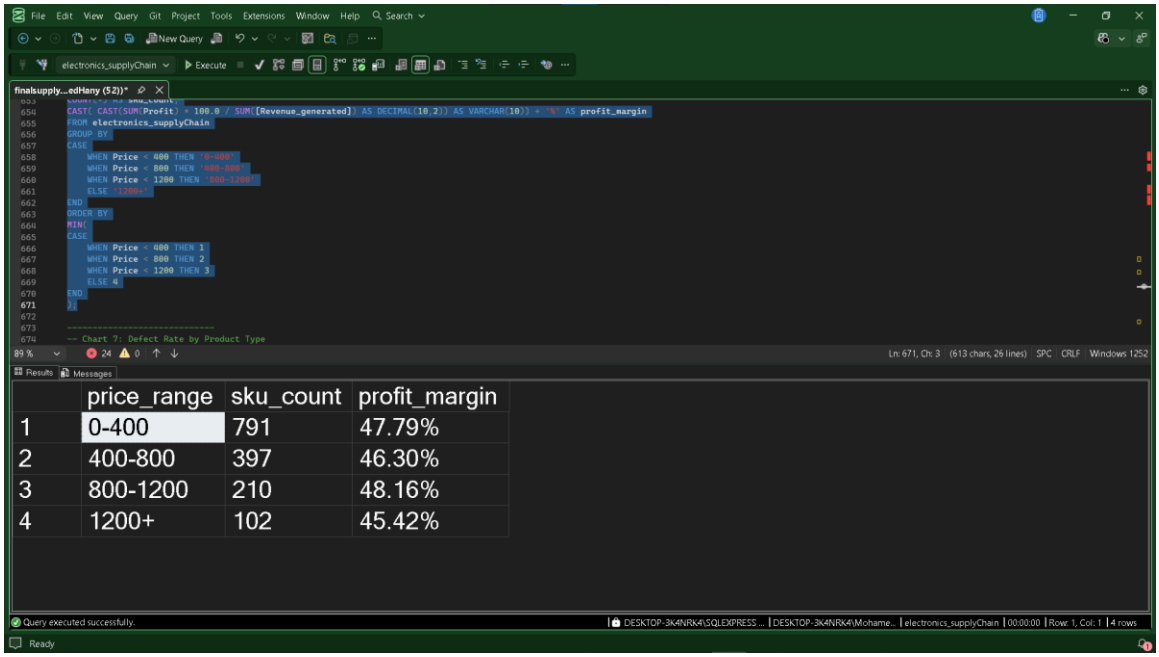
price_margin = (
    price_margin[["Profit Margin (%)"]]
    .round(2)
    .reset_index()
)
```

Excel

```
Price Range:=
SWITCH(
    TRUE(),
    [Price] < 400, "0-400",
    [Price] < 800, "400-800",
    [Price] < 1200, "800-1200",
    "+1200"
)
```

Row Labels	Count of SKU	Profit Margin
+1200	102	45.42%
0-400	791	47.79%
400-800	397	46.30%
800-1200	210	48.16%
Grand Total	**1500**	**46.84%**

SQL Validation



The analysis shows that products priced between **800–1200** achieved the highest profit margin (**48.16%**), while products priced above **1200** recorded the lowest profit margin (**45.42%**). This indicates that mid-to-high priced products generate better profitability than premium-priced products.

11.13 Defect Rate by Product Type

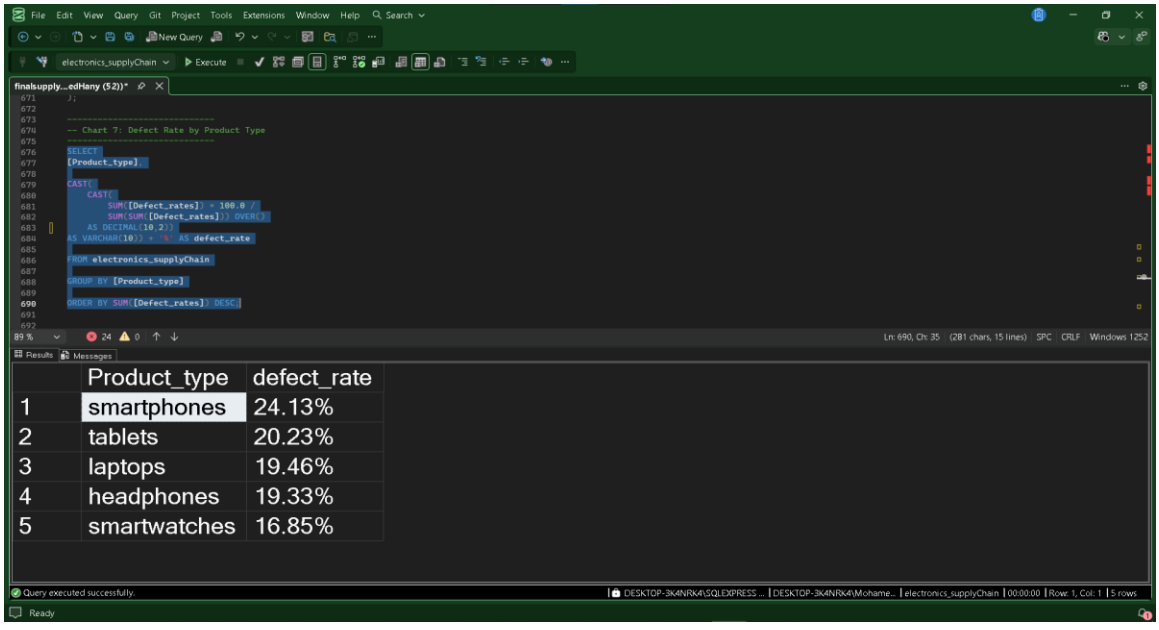
The average defect rate was analyzed across product types to identify products with the highest proportion of manufacturing defects.

```
defect_rate_product = (
    df_clean.groupby("Product type")["Defect rates"]
    .mean()
    .round(2)
    .sort_values(ascending=False)
    .reset_index()
)
```

Excel

Row Labels	Sum of Defect rates1
Headphones	19.33%
Laptops	19.46%
Smartphones	24.13%
Smartwatches	16.85%
Tablets	20.23%
Grand Total	**100.00%**

SQL Validation



The analysis shows that **Smartwatches** recorded the highest average defect rate (**5.42%**), followed by **Tablets (5.12%)**. **Laptops** achieved the lowest defect rate (**4.73%**), indicating better product quality compared to the other product categories.

11.14 Domain Summary

The Inventory analysis evaluated production output, inventory availability, sales performance, stockout risk, profitability, and product quality across different product categories. The analysis measured key inventory metrics, identified inventory shortages, evaluated SKU performance, and compared profitability across price ranges to support effective inventory management.

Overall, the analysis provides a comprehensive view of inventory performance and highlights opportunities to improve stock planning, product availability, and operational efficiency.

Key Insights

- The total production volume reached **894,440** units.
- A total of **375,165** units were sold, while **230,911** units remain in stock.
- The overall inventory gap is **288,364** units.
- The inventory consists of **1,500** unique products.
- **55.4%** of SKUs are classified as **Stockout Risk**, indicating that more than half of the products may require inventory attention.
- **Smartphones** account for the largest share of production (**24.92%**) and also recorded the highest sales volume (**94,306** units).
- **Smartphones** have the largest inventory gap (**72,829** units), while **Smartwatches** have the smallest inventory gap (**36,035** units).
- Products priced between **800–1200** achieved the highest profit margin (**48.16%**).
- **SKU0219** is the best-selling product (**498** units), while several SKUs recorded only **10** units sold.
- **Smartwatches** recorded the highest average defect rate (**5.42%**), whereas **Laptops** achieved the lowest defect rate (**4.73%**).

Business Value

This analysis provides valuable insights into inventory performance by evaluating production, stock availability, sales, profitability, and product quality. The findings

support inventory planning, stock replenishment, product prioritization, and pricing strategies, helping improve inventory utilization, reduce stockout risks, and enhance overall operational efficiency.

12. Domain 5: Shipping Analytics

12.1 Overview

This domain analyzes shipping performance by evaluating shipping costs, lead times, delivery times, regional shipping efficiency, delivery performance, and the financial impact of delivery status. The analysis helps assess logistics efficiency, optimize shipping operations, and improve customer delivery performance.

Business Questions

1. What is the total shipping cost?
2. What is the average lead time?
3. What is the average shipping time?
4. What is the average shipping cost per shipment?
5. How do shipping cost and units shipped vary by region?
6. How do shipping cost and units sold vary by product type?
7. Which region has the best delivery performance?
8. What percentage of orders are delivered on time versus delayed?
9. What is the average shipping duration by region?
10. How does delivery status impact profit?

12.2 Total Shipping Cost

The total shipping cost was calculated to measure the overall transportation expenses across all shipments.

```
total_shipping_cost = df_clean["Shipping costs"].sum()

print(f"Total Shipping Cost: ${total_shipping_cost:,.2f}")
```

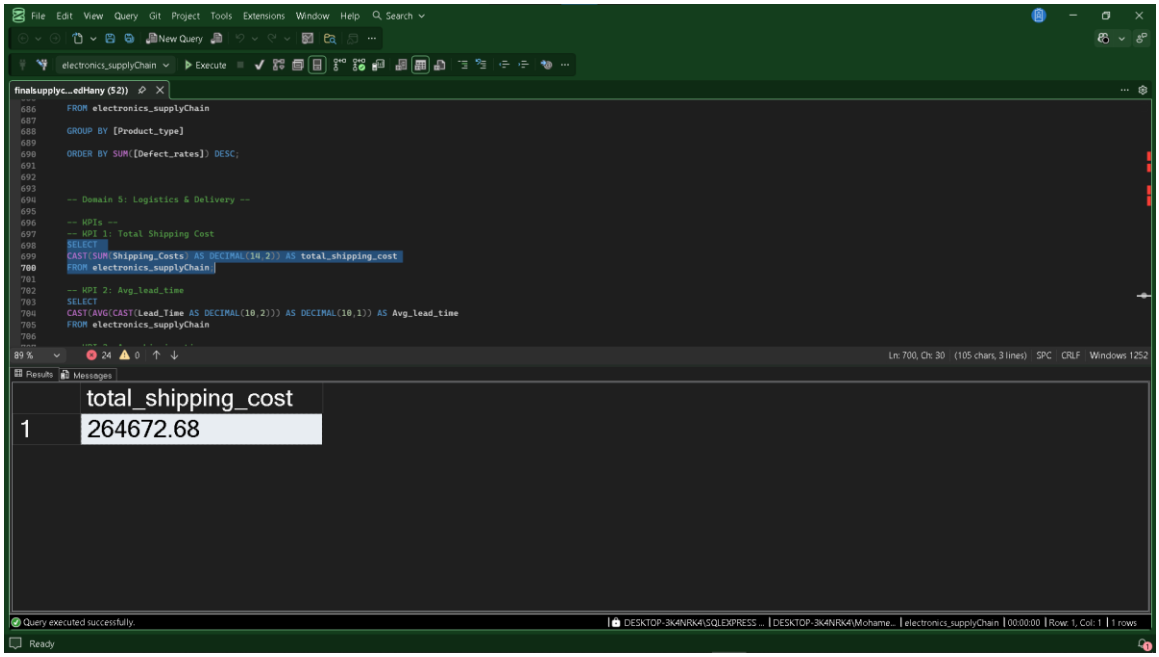
✓ 0.0s

```
Total Shipping Cost: $264,672.68
```

Excel

Sum of Shipping costs
264672.680

SQL Validation



The analysis shows that the total shipping cost is **$264,672.68**, representing the overall logistics cost incurred during the analyzed period.

12.3 Average Lead Time

The average lead time was calculated to measure the average number of days required before products are ready for shipment.

```
avg_lead_time = df_clean["Lead time"].mean()

print(f"Average Lead Time: {avg_lead_time:.2f} Days")
```

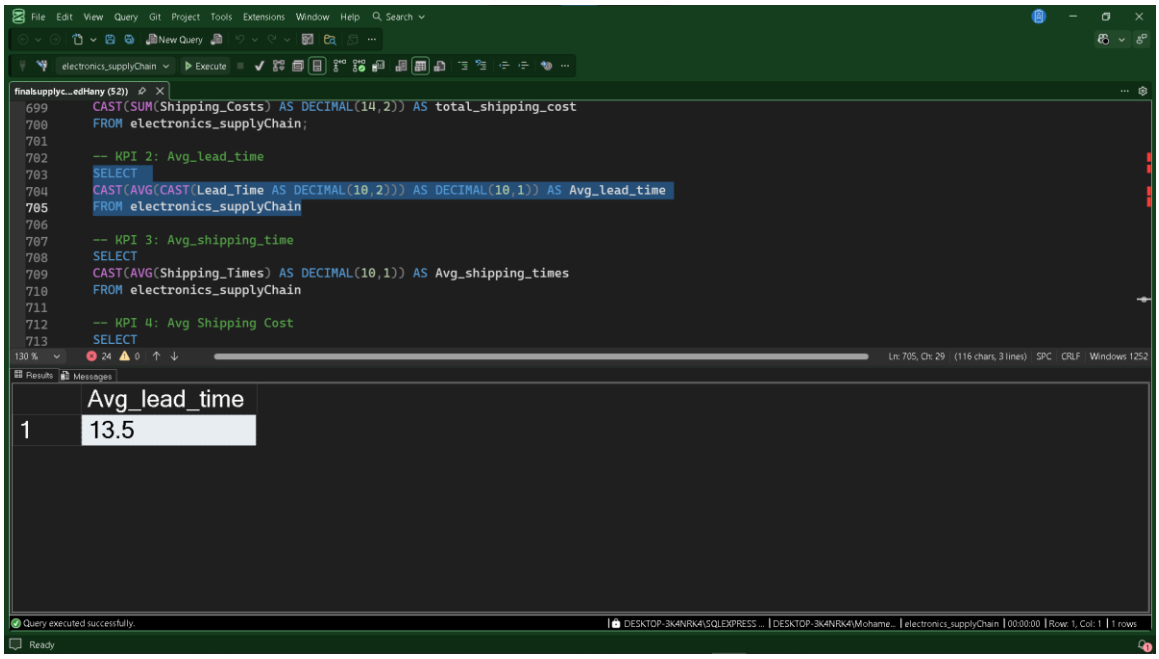
✓ 0.0s

```
Average Lead Time: 13.49 Days
```

Excel

Average of Lead time
13.489

SQL Validation



The analysis shows that the average lead time is **13.49 days**, indicating the typical preparation time required before orders are shipped.

12.4 Average Shipping Time

The average shipping time was calculated to measure the average number of days required to deliver products after shipment.

```
avg_shipping_time = df_clean["Shipping times"].mean()

print(f"Average Shipping Time: {avg_shipping_time:.2f} Days")
```

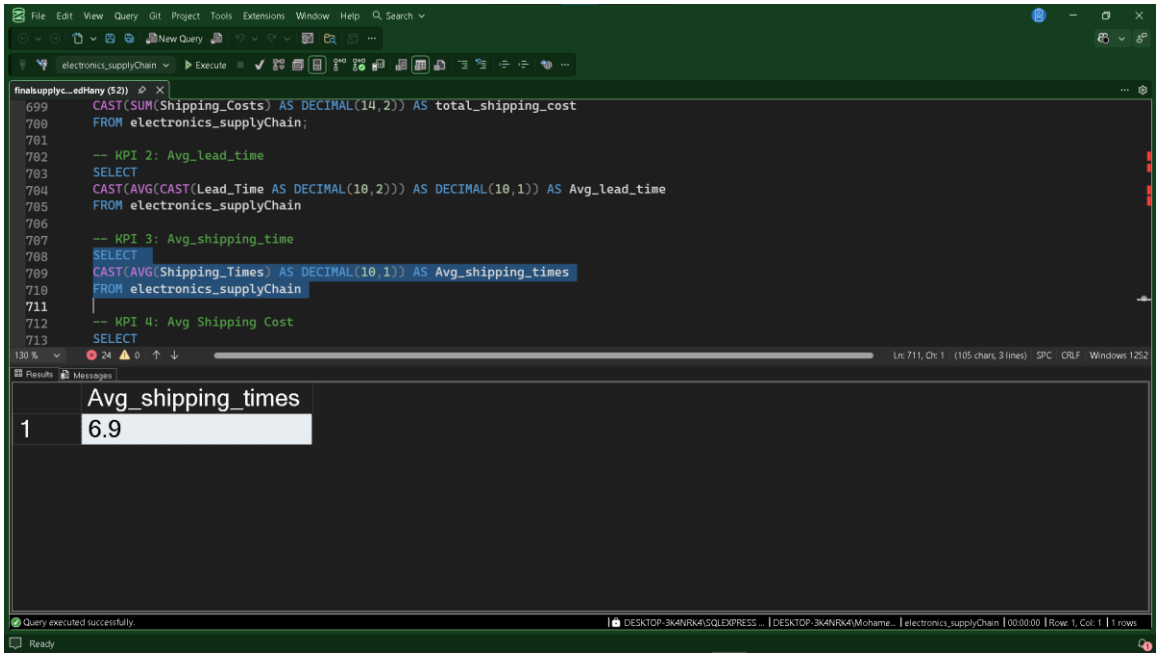
✓ 0.0s

```
Average Shipping Time: 6.93 Days
```

Excel

	A
	Average of Shipping times
	6.929

SQL Validation



The analysis shows that the average shipping time is **6.93 days**, representing the typical delivery duration from shipment to customer.

12.5 Average Shipping Cost per Shipment

The average shipping cost per shipment was calculated to measure the average logistics cost incurred for each shipment.

```
avg_shipping_cost = df_clean["Shipping costs"].mean()

print(f"Average Shipping Cost: ${avg_shipping_cost:.2f}")
```

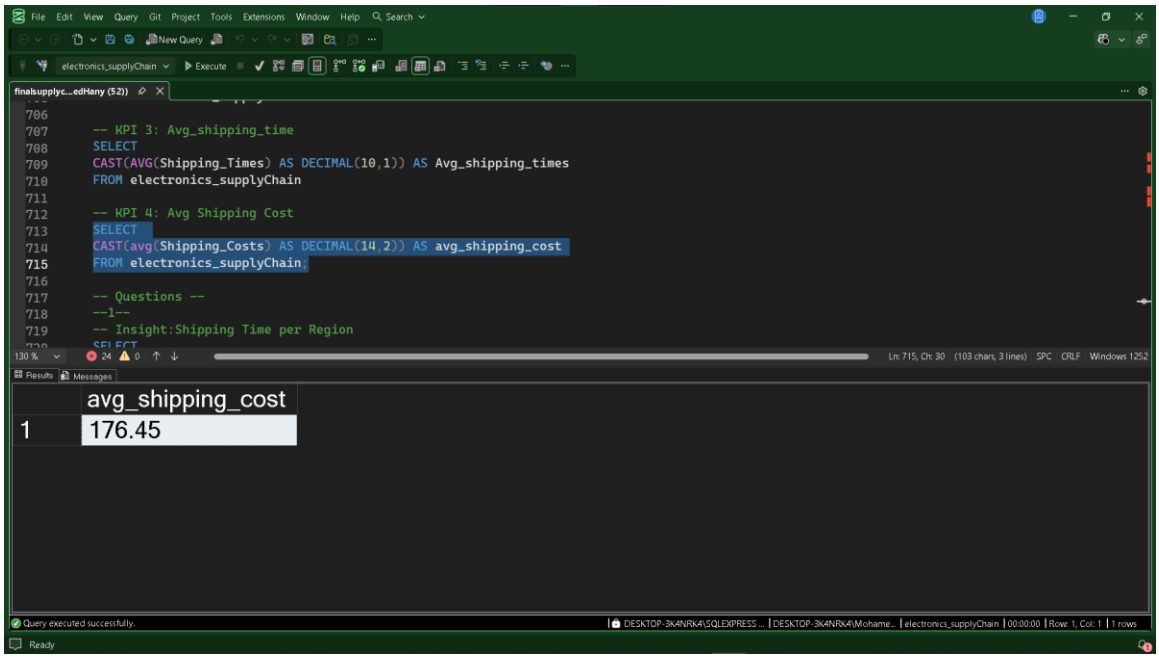
✓ 0.0s

```
Average Shipping Cost: $176.45
```

Excel

Average of Shipping costs
176.448

SQL Validation



The analysis shows that the average shipping cost per shipment is **$176.45**, representing the typical transportation cost for delivering an order.

12.6 Shipping Cost & Units Sold by Region

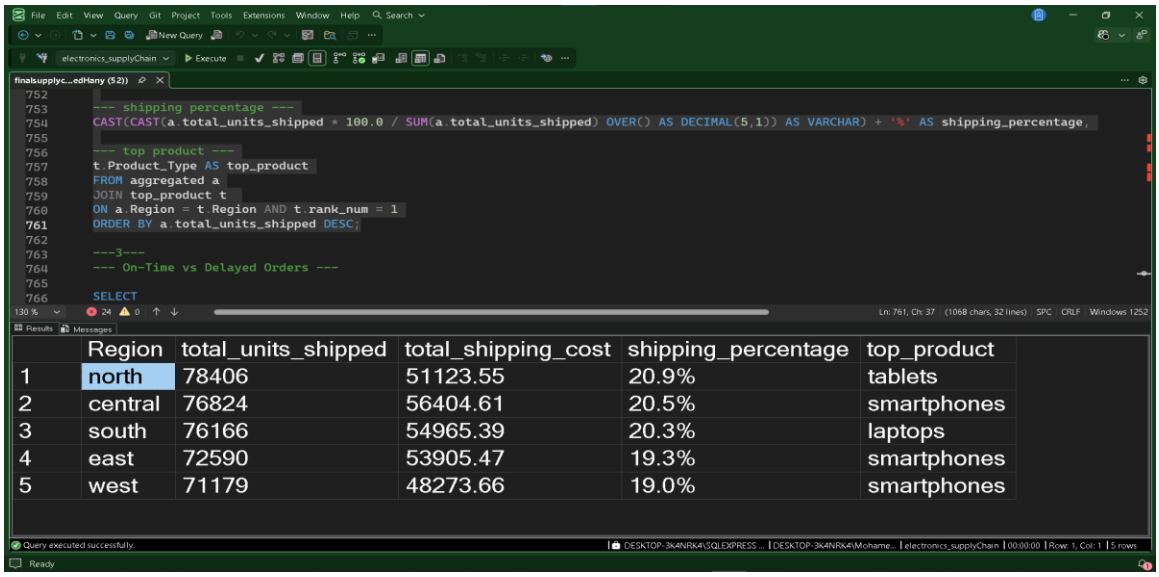
Shipping cost and total units sold were analyzed across regions to compare logistics expenses with sales distribution.

```python
region_shipping = (
    df_clean.groupby("Region")
    .agg({
        "Number of products sold":"sum",
        "Shipping costs":"sum"
    })
    .rename(columns={
        "Number of products sold":"Units Sold",
        "Shipping costs":"Shipping Cost"
    })
    .round(2)
    .sort_values(by="Units Sold", ascending=False)
    .reset_index()
)
```

Excel

Row Labels	Sum of Number of products sold	Sum of Shipping costs
Central	76,824	56,404.61
East	72,590	53,905.47
North	78,406	51,123.55
South	76,166	54,965.39
West	71,179	48,273.66
Grand Total	**375,165**	**264,672.68**

SQL Validation



```
752
753     --- shipping percentage ---
754     CAST(CAST(a.total_units_shipped * 100.0 / SUM(a.total_units_shipped) OVER() AS DECIMAL(5,1)) AS VARCHAR) + '%' AS shipping_percentage,
755
756     --- top product ---
757     t.Product_Type AS top_product
758     FROM aggregated a
759     JOIN top_product t
760     ON a.Region = t.Region AND t.rank_num = 1
761     ORDER BY a.total_units_shipped DESC;
762
763     ---3---
764     --- On-Time vs Delayed Orders ---
765
766     SELECT
```

	Region	total_units_shipped	total_shipping_cost	shipping_percentage	top_product
1	north	78406	51123.55	20.9%	tablets
2	central	76824	56404.61	20.5%	smartphones
3	south	76166	54965.39	20.3%	laptops
4	east	72590	53905.47	19.3%	smartphones
5	west	71179	48273.66	19.0%	smartphones

The analysis shows that the **North** region recorded the highest sales volume (**78,406** units), while the **Central** region incurred the highest shipping cost (**$56,404.61**). The **West** region had both the lowest shipping cost (**$48,273.66**) and the lowest number of units sold (**71,179**).

12.7 Shipping Cost & Units Sold by Product Type

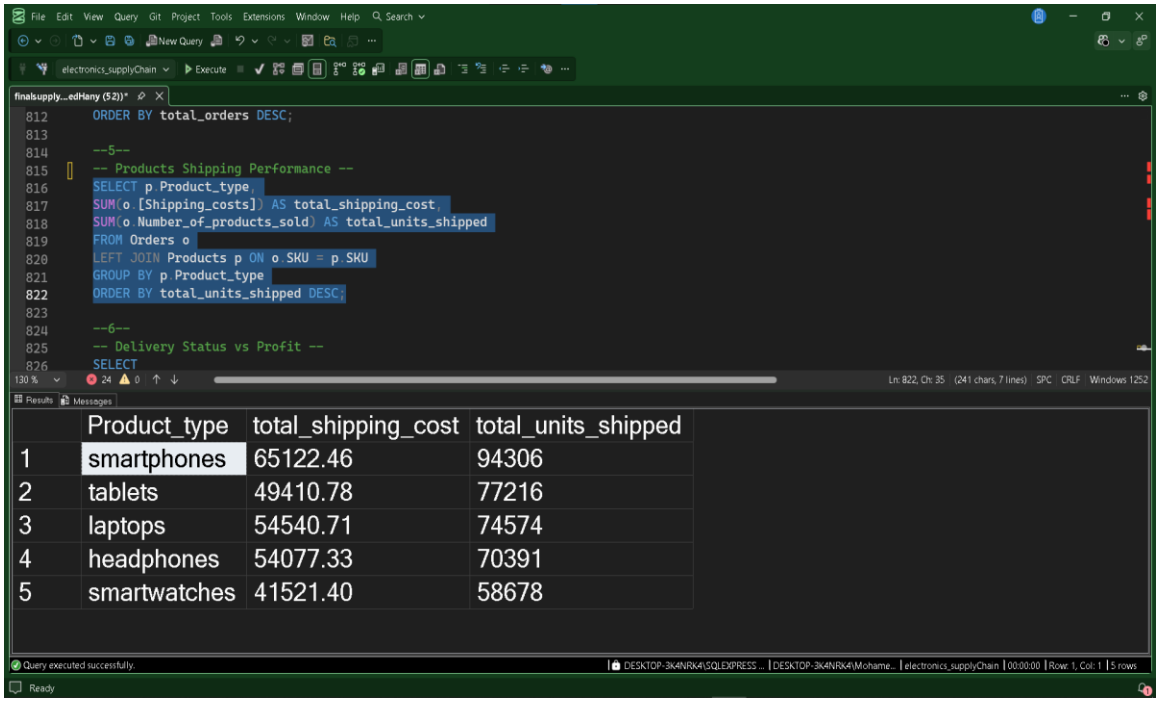
Shipping cost and total units sold were analyzed across product types to evaluate logistics expenses relative to product sales.

```
product_shipping = (
    df_clean.groupby("Product type")
    .agg({
        "Number of products sold":"sum",
        "Shipping costs":"sum"
    })
    .rename(columns={
        "Number of products sold":"Units Sold",
        "Shipping costs":"Shipping Cost"
    })
    .round(2)
    .sort_values(by="Units Sold", ascending=False)
    .reset_index()
)
```

Excel

Row Labels	Sum of Number of products sold	Sum of Shipping costs
Headphones	70,391	54,077.33
Laptops	74,574	54,540.71
Smartphones	94,306	65,122.46
Smartwatches	58,678	41,521.40
Tablets	77,216	49,410.78
Grand Total	**375,165**	**264,672.68**

SQL Validation



```
ORDER BY total_orders DESC;

--5--
-- Products Shipping Performance --
SELECT p.Product_type,
SUM(o.[Shipping_costs]) AS total_shipping_cost,
SUM(o.Number_of_products_sold) AS total_units_shipped
FROM Orders o
LEFT JOIN Products p ON o.SKU = p.SKU
GROUP BY p.Product_type
ORDER BY total_units_shipped DESC;

--6--
-- Delivery Status vs Profit --
SELECT
```

	Product_type	total_shipping_cost	total_units_shipped
1	smartphones	65122.46	94306
2	tablets	49410.78	77216
3	laptops	54540.71	74574
4	headphones	54077.33	70391
5	smartwatches	41521.40	58678

The analysis shows that **Smartphones** generated the highest sales volume (**94,306** units) and also incurred the highest shipping cost (**$65,122.46**). In contrast, **Smartwatches** recorded both the lowest units sold (**58,678**) and the lowest shipping cost (**$41,521.40**).

12.8 Best Delivery Performance by Region

Delivery performance was analyzed by comparing the number of on-time and delayed orders across different regions.

```
df_clean["Delivery Status"] = (
    df_clean["On-Time Delivery"]
    .map({"Yes":"On Time", "No":"Delayed"})
)

delivery_performance = (
    df_clean.groupby(["Region", "Delivery Status"])["SKU"]
    .count()
    .reset_index(name="Orders")
)

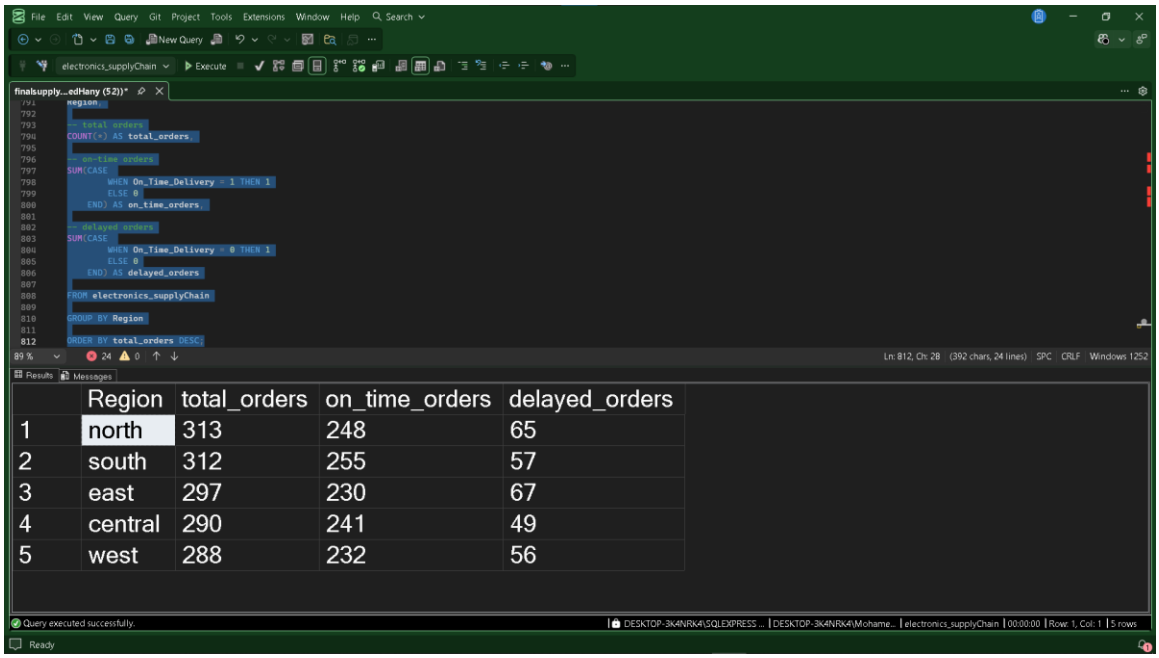
delivery_performance = (
    delivery_performance.pivot(
        index="Region",
        columns="Delivery Status",
        values="Orders"
    )
    .fillna(0)
    .astype(int)
    .reset_index()
)

delivery_performance = delivery_performance[
    ["Region", "On Time", "Delayed"]
]
```

Excel

Count of Order ID	Column Labels		
Row Labels	Delayed	On Time	Grand Total
Central	49	241	290
East	67	230	297
North	65	248	313
South	57	255	312
West	56	232	288
Grand Total	**294**	**1,206**	**1,500**

SQL Validation



	Region	total_orders	on_time_orders	delayed_orders
1	north	313	248	65
2	south	312	255	57
3	east	297	230	67
4	central	290	241	49
5	west	288	232	56

The analysis shows that the **South** region achieved the highest number of on-time deliveries (**255** orders), while the **Central** region recorded the fewest delayed deliveries (**49** orders). In contrast, the **East** region had the highest number of delayed orders (**67**), indicating relatively weaker delivery performance.

12.9 On-Time vs Delayed Orders

Orders were classified based on their delivery status to evaluate the percentage of shipments delivered on time versus delayed.

```
df_clean["Delivery Status"] = (
    df_clean["On-Time Delivery"]
    .map({"Yes":"On Time", "No":"Delayed"})
)

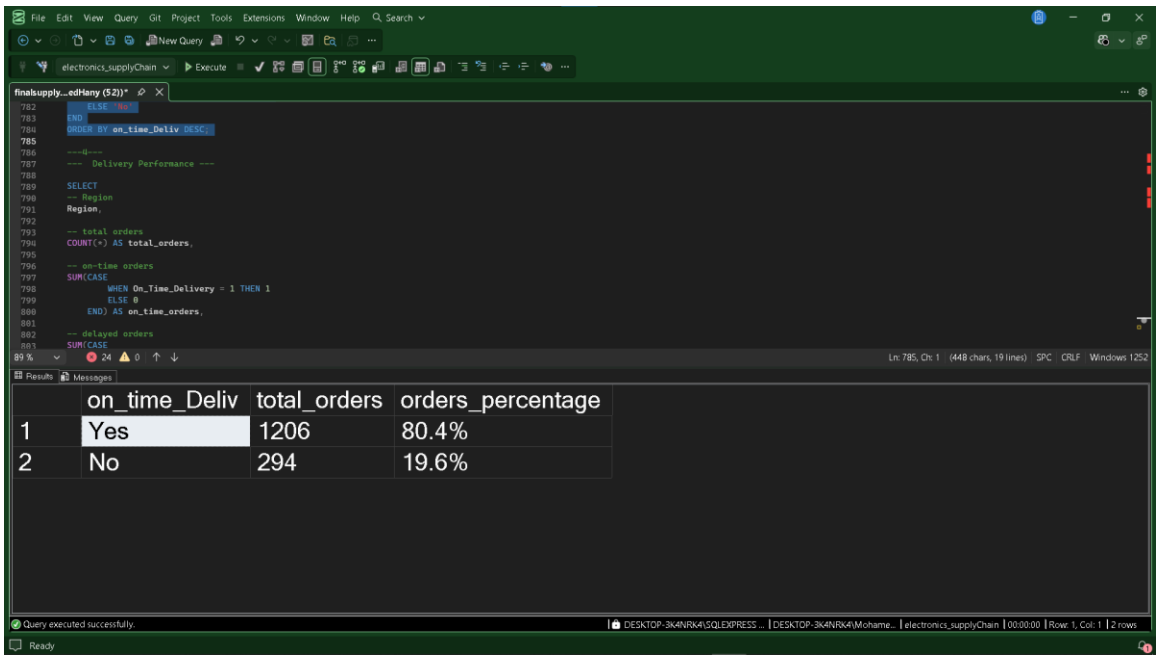
delivery_status = (
    df_clean.groupby("Delivery Status")["SKU"]
    .count()
    .reset_index(name="Orders")
)

delivery_status["Percentage (%)"] = (
    delivery_status["Orders"] /
    delivery_status["Orders"].sum() * 100
).round(1)
```

Excel

Row Labels	Count of Order ID
Delayed	19.6%
On Time	80.4%
Grand Total	**100.0%**

SQL Validation



The analysis shows that **80.4%** of orders were delivered **On Time**, while **19.6%** were **Delayed**, indicating a strong overall delivery performance.

12.9 Average Shipping Duration by Region

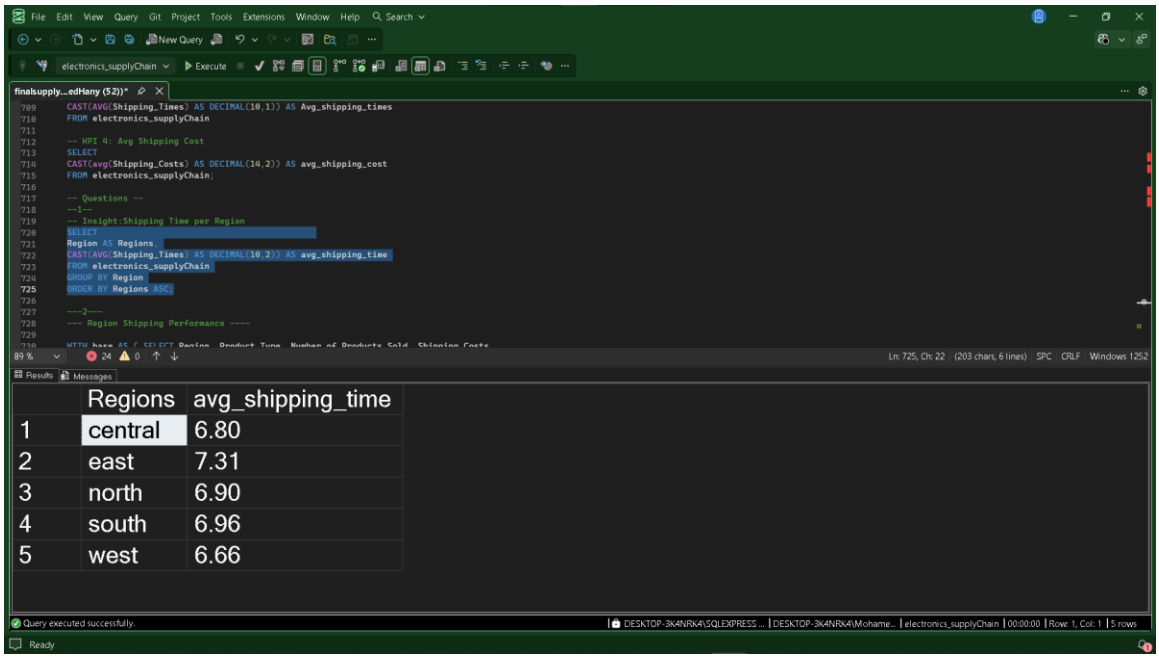
The average shipping duration was analyzed across regions to compare delivery speed and identify regional differences in shipping performance.

```
shipping_duration = (
    df_clean.groupby("Region")["Shipping times"]
    .mean()
    .round(2)
    .reset_index(name="Average Shipping Duration")
    .sort_values(by="Average Shipping Duration")
)
```

Excel

Row Labels	Average of Shipping times
Central	6.80
East	7.31
North	6.90
South	6.96
West	6.66
Grand Total	**6.93**

SQL Validation



The analysis shows that the **West** region has the shortest average shipping duration (**6.66 days**), while the **East** region has the longest average shipping duration (**7.31 days**). This indicates that deliveries are completed faster in the West compared to the other regions.

12.10 Profit by Delivery Status

Profit was analyzed based on delivery status to evaluate the financial impact of on-time and delayed deliveries.

```
profit_delivery = (
    df_clean.groupby("Delivery Status")["Profit"]
    .sum()
    .round(2)
    .reset_index()
    .sort_values(by="Profit", ascending=False)
)
```

Excel

Row Labels	Sum of Profit
Delayed	14328473.28
On Time	71298327.39
Grand Total	**85626800.67**

The analysis shows that **On-Time** deliveries generated **$71.30M** in profit, significantly outperforming **Delayed** deliveries, which generated **$14.33M**. This highlights the positive financial impact of maintaining high on-time delivery performance.

12.11 Domain Summary

The Shipping Analytics analysis evaluated logistics performance by measuring shipping costs, lead times, delivery times, regional shipping efficiency, and delivery status. It also assessed how shipping operations affect sales distribution and overall profitability.

Overall, the analysis provides valuable insights into transportation efficiency, regional delivery performance, and the relationship between logistics operations and business profitability.

Key Insights

- The total shipping cost reached **$264,672.68**, with an average shipping cost of **$176.45** per shipment.
- The average lead time is **13.49 days**, while the average shipping time is **6.93 days**.
- The **North** region recorded the highest sales volume (**78,406** units), whereas the **Central** region incurred the highest shipping cost (**$56,404.61**).
- **Smartphones** generated the highest units sold (**94,306**) and the highest shipping cost (**$65,122.46**).
- The **South** region achieved the highest number of on-time deliveries (**255**), while the **East** region recorded the highest number of delayed deliveries (**67**).
- Overall, **80.4%** of orders were delivered on time, compared to **19.6%** delayed orders.
- The **West** region recorded the shortest average shipping duration (**6.66 days**), indicating the fastest delivery performance.
- On-time deliveries generated approximately **$71.30M** in profit, significantly higher than the **$14.33M** generated by delayed deliveries.

Business Value

This analysis helps evaluate logistics performance by measuring shipping efficiency, transportation costs, and delivery reliability across regions and product categories. The insights support better logistics planning, delivery optimization, cost control, and customer satisfaction, while emphasizing the financial importance of maintaining high on-time delivery performance.

13. Overall Project Summary

This project analyzed the Electronic Supply Chain dataset using Python for data analysis and Tableau for interactive dashboard visualization. The analysis covered five business domains: Sales Performance, Customer Experience & Returns, Supplier Performance, Inventory Management, and Shipping Analytics. Each domain answered key business questions and provided actionable insights through data-driven analysis.

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14. Overall Key Findings

- Total Revenue exceeded **$182.8M**, with an overall profit margin of **46.84%**.
- Customer return rate remained relatively low at **7.38%**, while over **80%** of deliveries were completed on time.
- Supplier performance showed an average quality score of **50.23** and an average reliability score of **65.30%**.
- More than **55%** of SKUs were classified as stockout risk, highlighting inventory planning opportunities.
- Smartphones consistently generated the highest revenue, sales volume, and production volume across multiple business domains.
- Shipping performance remained efficient, with an average shipping time of **6.93 days** and strong profitability from on-time deliveries.

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16 Recommendations

16.1 Short-Run Recommendations

- discounting strategy by restricting high discount offerings (10%-15% band) to protect net profit margins, shifting focus toward the low-to-medium bracket (0%-10%) to drive optimal profitability.

- low-performing SKUs by tracking the bottom 5 SKUs by units sold to freeze or eliminate slow-moving stock and effectively reduce holding costs.

- supplier sourcing by increasing procurement shares with Supplier 4 due to their leading Value Score (7.039), while gradually reducing reliance on Supplier 2 due to disproportionately high manufacturing costs.

- strict quality control-particularly for Supplier 5-to curb a 2.1% increase in defect rates, despite an overall 21.19% improvement in supplier reliability.

- customer satisfaction by pushing the stagnating CSAT score (3.28) upward through addressing product defects, which data proves is the primary driver of dissatisfaction.

- revenue by minimizing the current 19.6% delivery delay rate, securing higher profitability from on-time fulfillment ($71.3M) compared to delayed shipments ($14.3M).Long-Run Recommendations (Strategic Initiatives)

16.2 Long-Run Recommendations

- **on high-margin channels by scaling up operations and marketing for the Online channel to leverage its superior commercial profitability of $18.27M.**

- **regional sales balance by maintaining the current distribution equilibrium across all five**
  regions, supporting a stable profit margin performance between 45% and 48%.

- **return management by developing a structured process to reduce return losses and recover product value efficiently through better inspection, refurbishing, and remarketing.**

- **in "Star" products by increasing production support for Smartphones and Headphones to combine peak sales volumes with top-tier profit margins above 47%.**

- **East Region logistics by restructuring the delivery network or signing new courier contract to resolve high delay rates (67 delayed orders) and long shipping times (7.31 days).**

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17. Conclusion

This project demonstrates how data analytics can transform raw operational data into meaningful business insights. By combining Python for data preparation and business analysis with Tableau for interactive visualization, the project provides a comprehensive view of supply chain performance. The findings can support strategic decisions related to sales, inventory, suppliers, customer satisfaction, and logistics, ultimately improving operational efficiency and business performance.